

Quantile and Distributional Difference-in-Differences: A Survey and Practitioner’s Guide

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Abstract

This paper is a survey and practitioner guide to quantile and distributional difference-in-differences (QDiD). The main object of interest is the treated group’s missing post-treatment untreated distribution. Changes-in-changes constructs it through rank transport; distributional parallel trends restricts cumulative probabilities at outcome thresholds; panel methods add restrictions on transitions and dependence; staggered designs build cohort-time distributions; and partial-identification methods retain a set when no point restriction is credible. The discussion follows each method from its economic argument to the way its counterfactual distribution is estimated, diagnosed, and used for inference. A running application and an assumption focused Monte Carlo show why model agreement is not identification and why resampling must follow treatment assignment. The practical lesson is to define the distributional target first, defend the counterfactual restriction, show evidence tied to it, and report sensitivity or bounds when the data do not select a point-identifying method.

1 Introduction

Difference-in-differences (DiD) is used to estimate an average treatment effect. Many policy questions, however, concern other features of the outcome distribution. A minimum-wage increase may affect unemployment or wages differently near the bottom and the top of the distribution. A health intervention may primarily reduce the probability of a very poor outcome. A training program may help workers with low counterfactual earnings while doing little at the median. Quantile and distributional DiD methods are designed for these questions. The broader treatment-effects literature has long emphasized that mean effects can conceal economically important heterogeneity and distributional changes (Doksum, 1974; Heckman et al., 1997; Bitler et al., 2006).

The literature offers several ways to estimate distributional effects, but their assumptions are not interchangeable. The common missing object is the distribution that treated units would have experienced after treatment had they remained untreated. This paper is a survey and practitioner guide about different method of estimating counterfactual distribution. Its purpose is to make the existing methods easier to compare and use. The core review includes peer-reviewed articles and recent working papers that identify a DiD counterfactual distribution, change the data or dependence restrictions needed for that object, or supply directly relevant inference. Working papers are treated as evolving contributions. Papers concerned only with mean effects, generic cross-sectional quantile treatment effects, or applications without a methodological QDiD contribution are used only for context; Appendix A gives the auditable inclusion map. An

earlier broad review is provided by Lechner (2011). This paper begins the discussion with a canonical two-group, two-period model and organizes later methods by which element of that model changes. I begins with the target parameter and identifying assumptions rather than with a preferred regression, use one running application, and keeps estimation choices tied to the underlying research design.

The canonical QDiD problem is easy to state. The treated post-treatment distribution is observed; the treated post-treatment distribution without treatment is not. Once that missing distribution is constructed, researchers can report effects on cumulative probabilities, quantiles, inequality measures, or tail risks. The difficult step is therefore not quantile estimation itself. It is choosing and defending the restriction used to construct the counterfactual distribution.

The survey uses that observation to organize the literature. As the discussion moves from one method family to the next, it returns to the same underlying issues: the target, the available data, the restriction that constructs the counterfactual distribution, the evidence bearing on that restriction, and the uncertainty that remains. Following these connections makes clear which methods are substitutes, which require additional panel or timing information, and which change the target population.

QDiD combines three literatures that are often discussed separately. The first defines and estimates quantile and distributional treatment effects (Doksum, 1974; Koenker and Bassett, 1978; Abadie et al., 2002; Firpo, 2007; Donald and Hsu, 2014). The second constructs and conducts inference on counterfactual distributions and decompositions (DiNardo et al., 1996; Machado and Mata, 2005; Rothe, 2010; Chernozhukov et al., 2013; Firpo et al., 2011). The third studies identification, covariate adjustment, diagnostics, and aggregation in DiD (Abadie, 2005; Sant’Anna and Zhao, 2020; Callaway and Sant’Anna, 2021; Goodman-Bacon, 2021; Sun and Abraham, 2021; Borusyak et al., 2024). These adjacent results supply targets, inferential tools, and design lessons, but they become QDiD methods only when paired with an explicit restriction for the treated group’s missing post-treatment untreated distribution. A related but distinct literature builds a missing counterfactual distribution from a weighted combination of donor units rather than from a two-group parallel-trends argument, most notably Gunsilius (2023); where credible donor pools exist, distributional synthetic controls are a genuine alternative to the DiD restrictions surveyed here, not a special case of them.

Two empirical exercises support the guide. First, a Monte Carlo assumption laboratory illustrates how estimators behave when their own identifying restriction holds, when a competing restriction holds, and when neither holds. These simulations do not rank methods universally; the data-generating processes are intentionally chosen to clarify the consequences of alternative assumptions. Second, a minimum-wage illustration applies several methods to the same counties and target quantiles. It reports state-clustered uniform bands, placebo processes, dependence-robust bounds, and leave-one-state-out results. The application is deliberately allowed to end without a preferred point estimate.

The paper’s practical message is that method choice should be visible. Researchers should define the estimand, state the counterfactual restriction, show diagnostics matched to that restriction, report the full estimated counterfactual distribution before selected quantiles, and distinguish sampling uncertainty from uncertainty about identification. When no point-identifying story is persuasive, bounds or an explicit decision not to report a point estimate are legitimate outcomes.

The remainder of the paper follows the canonical-model-and-relaxations structure used in the benchmark surveys. After the running example, two sections define the canonical QDiD problem and compare alternative counterfactual constructions. Five method-family sections then move through rank-based, panel, distributional-trend, staggered, and specialized extensions, explaining how identification becomes an estimator in each case. The final sections connect software and diagnostics to a minimum-wage illustration and simulation evidence, and then discuss how the pieces can be presented in an applied paper.

2 Running Example: Minimum Wages and Distributional Effects

Consider a state-level minimum-wage increase. A conventional DiD analysis asks whether average wages or average employment changed in treated states relative to comparison states. A QDiD analysis asks richer questions: Did the reform raise the 10th percentile of wages? Did it compress the wage distribution? Did the lower tail of hours worked fall? Did any employment losses concentrate among workers who would otherwise have been near the bottom of the wage distribution?

The example clarifies why different QDiD papers impose different assumptions. If the comparison states provide a credible distributional time trend, distributional parallel trends is natural. If workers preserve latent ranks as the wage schedule changes, changes-in-changes is natural. If linked panel data reveal how workers move through the distribution over time, copula or dependence restrictions become more plausible. If states adopt minimum-wage changes at different dates, the relevant primitive is a collection of cohort-time distributional effects rather than one overall QTT. Section 11 returns to this example and applies the competing restrictions, diagnostics, and bounds to the same data.

3 The Canonical Two-Group, Two-Period QDiD Setup

Consider a two-group, two-period design with group $G \in \{0, 1\}$ and period $T \in \{0, 1\}$. Group $G = 1$ is treated in the post period; group $G = 0$ is untreated in both periods. Let $Y_t(1)$ and $Y_t(0)$ denote potential outcomes at time t with and without treatment. The observed outcome is $Y_t = Y_t(0)$ except for treated units in the post period, for whom $Y_1 = Y_1(1)$.

The average treatment effect on the treated is

$$ATT = E[Y_1(1) - Y_1(0) \mid G = 1].$$

The observed component is $E[Y_1(1) \mid G = 1]$. The missing object is $E[Y_1(0) \mid G = 1]$.

However, QDTT and DTT replaces this missing mean problem with the missing distribution problem.

$$F_{1,1}^0(y) = \Pr(Y_1(0) \leq y \mid G = 1).$$

Once this distribution is identified, many estimands follow.

The distributional treatment effect at threshold y is

$$DTT(y) = F_{Y_1(1)|G=1}(y) - F_{Y_1(0)|G=1}(y).$$

It measures how treatment changes the probability of falling below a fixed outcome threshold. Under this sign convention, a positive DTT means more probability mass below y ; whether that is beneficial depends on the outcome. In the minimum-wage example, this might be the effect on the probability that wages are below a poverty-relevant threshold. Every empirical DTT figure should restate this sign convention.

The quantile treatment effect on the treated is

$$QTT(\tau) = Q_{Y_1(1)|G=1}(\tau) - Q_{Y_1(0)|G=1}(\tau), \quad \tau \in (0, 1),$$

where $Q_Z(\tau) = \inf\{y : F_Z(y) \geq \tau\}$. Differences of marginal quantiles originate in the statistical treatment-effect literature and are distinct from conditional quantile-regression coefficients (Doksum, 1974; Koenker and Bassett, 1978; Koenker, 2005; Angrist et al., 2006). A QTT compares the treated outcome distribution with the counterfactual untreated distribution at the same marginal quantile. However, QTT is not generally the τ th quantile of $Y_1(1) - Y_1(0)$ for treated units. The equality requires additional assumptions about the joint distribution of treated and untreated potential outcomes, such as rank invariance or rank similarity. The same distinction motivates the literature on bounds for the distribution of individual gains and on local quantile effects under instrumental variables (Heckman et al., 1997; Abadie et al., 2002; Fan and Park, 2010). This distinction is central for interpretation. A positive QTT at the 10th percentile means that the 10th percentile of the treated marginal distribution is higher than the 10th percentile of the counterfactual marginal distribution. It does not necessarily mean that individuals at the 10th percentile of the untreated distribution experienced that gain. Table 1 explains common notations and terminologies used in this paper.

Table 1: Common notation and terminology

Symbol or term	Meaning used throughout
F_{11}^0, F_{11}^1	Untreated counterfactual and treated observed post-period CDFs for the treated group
$DTT(y)$	Difference $F_{11}^1(y) - F_{11}^0(y)$ at a fixed outcome threshold (y)
$QTT(\tau)$	Difference of marginal quantiles $Q_{11}^1(\tau) - Q_{11}^0(\tau)$
$CQTT(\tau x)$	QTT conditional on covariates $X = x$
$F_{g,t}^0, QTT_{g,t}(\tau)$	Cohort-time counterfactual CDF and QTT in staggered designs
QDiD	Umbrella term for quantile and distributional DiD designs
DPT	Distributional parallel trends, the threshold-by-threshold identifying restriction
DDiD	Estimator/model family that operationalizes a distributional DiD restriction
CiC; distribution-regression DiD	Changes-in-changes; link-scale regression for the distributional process
Quantile DiD	Additive change in marginal quantiles at a common rank; distinct from DPT
Copula/dependence restriction	A rule coupling untreated baseline ranks with untreated-change ranks; stability may be imposed over time or across groups

4 Counterfactual Constructions and Method Choice

Section 3 reduced the QDiD problem to one missing object, $F_{Y_1(0)|G=1}$, and to one warning: a QTT is a difference of marginal quantiles, not generally the quantile of an individual gain. What it did not yet say is

that the literature offers several genuinely different ways to recover that missing distribution, and that the choice among them is the most consequential decision in a QDiD application – more consequential than any later choice of estimator, bandwidth, or bootstrap scheme. Canonical DiD restricts only the mean of the missing distribution,

$$E[Y_1(0) - Y_0(0) \mid G = 1] = E[Y_1(0) - Y_0(0) \mid G = 0],$$

which is too weak for a QTT because it constrains a single functional of an entire distribution. Distributional parallel trends (DPT) strengthens this restriction to hold at every outcome threshold. Changes-in-changes (CiC) asks a different question: not whether cumulative probabilities move in parallel, but whether untreated units keep a stable latent rank as the outcome schedule shifts. Panel and repeated-cross-section copula restrictions go further still, using information about how units move through the distribution over time to couple baseline levels with subsequent changes. And when none of these stories is credible, partial-identification methods retain a set rather than resolve the missing dependence by assumption. Sections 5 through 8 take up these four families one at a time, each moving from its economic argument to an estimator, its diagnostics, and its characteristic failure mode.

Before turning to the details, it is worth being precise about bounds, because they are a first-class design choice and not a fallback for a failed point estimate (Heckman et al., 1997; Fan and Park, 2010; Rothe, 2012). If observed-data and shape restrictions imply $L(y) \leq F_{11}^0(y) \leq U(y)$ for every y , quantile inversion reverses the ordering and the identified set for the QTT is

$$\mathcal{I}_{QTT}(\tau) = [Q_{11}^1(\tau) - Q_L(\tau), Q_{11}^1(\tau) - Q_U(\tau)].$$

The panel bounds of Fan and Yu (2012), used again in Section 11, make this concrete: assuming overlap and equal distributions of untreated outcome changes across treated and comparison groups, Makarov–Fréchet bounds on their sum give sharp CDF envelopes for $Y_1(0) \mid G = 1$, and inversion gives sharp bounds on the marginal QTT – not on the quantiles of individual gains. A credible bounds analysis reports the assumptions behind L and U , the bound width over the target quantile range, and which substantive conclusions hold at every point in the set; sampling uncertainty attaches to the set as a whole, not to a chosen endpoint. A sensitivity path between two point-identifying models, such as $F_{11,\lambda}^0 = (1-\lambda)F_{11}^{0,CiC} + \lambda F_{11}^{0,DPT}$ for $\lambda \in [0, 1]$, is a different and weaker exercise: it shows how a conclusion moves across a chosen family of models, and it becomes an identified set only when every admissible counterfactual is known to lie inside the reported range.

4.1 Choosing a starting method

The data structure available narrows the feasible methods without selecting one automatically: panel histories make transition restrictions available, repeated cross-sections favor marginal-distribution restrictions, staggered timing requires cohort–time targets, and specialized tail, compliance, or multivariate estimands require additional structure of their own. Table 2 turns this into a working crosswalk for the four main

families, recording for each one the data it needs, the restriction it imposes, the observable implication that speaks to that restriction, and what remains unidentified even when the restriction holds. Table 3 does the same for the designs that change the target or weaken the conclusion rather than merely changing the estimator – staggered adoption, extreme tails, fuzzy compliance, and the case where only support or shape restrictions are credible. Neither table is an algorithm: it names a starting point that substantive knowledge about the setting must still justify, and agreement between two point estimators built on different restrictions is descriptive sensitivity evidence, not proof that either restriction is correct or a narrowing of a formal identified set.

Table 2: Assumption crosswalk for the main QDiD method families

Family	ID result	Feasible data	Counterfactual restriction	Observable implication/check	Support/shape burden	Still not identified	Fallback
CiC	Point	Panel or repeated CS	Stable scalar ranks and common monotone untreated change	Support, composition, placebo transports	Continuous canonical outcome; treated ranks inside control support	Individual gain distribution without rank coupling	DPT sensitivity; bounds
DPT	Point	Panel or repeated CS	Parallel untreated CDF evolution at every threshold	Threshold pretrends; raw implied-CDF range and monotonicity	Common threshold support; valid CDF after sampling uncertainty	Individual gains; postperiod validity from pretrend non-rejection	CiC sensitivity; bounds
Additive quantile DiD	Point	Panel or repeated CS	Parallel untreated changes in marginal quantiles at a common rank	Quantile pretrends; rank-grid sensitivity	Quantile support and stable inversion	Threshold-DPT counterfactual; individual gains	DPT/CiC comparison; bounds
Panel QTT	Point	Linked panel with sufficient periods	Stable untreated-change marginal plus stated baseline-change dependence	Pretreatment transitions, dependence stability, attrition	Baseline/change support; copula extrapolation	Counterfactual coupling absent its restriction	Weaker copula class; Makarov bounds
Conditional QDiD with nuisance DR	Point under the chosen base restriction	Panel or repeated CS with X	CiC, DPT, or another QDiD restriction holds conditional on X	Overlap, weight tails, both nuisance fits	Treated covariate overlap; valid conditional CDF	The causal target when the base restriction fails; consistency when both nuisance models fail	Simpler nuisance models; narrower target; bounds
Distribution-regression DiD	Point on the stated link scale	Repeated CS or panel; uni-/multivariate outcomes	No untreated group-time interaction on the chosen distribution-regression link; joint dependence model if multivariate	Link sensitivity, marginal fit, raw joint-CDF validity	Threshold/covariate support; link and copula extrapolation	Joint effect from marginal trends alone	Alternative links; marginal effects; bounds

Reading rule: Begin with the counterfactual restriction, then verify that the command implements that operator. DPT and additive quantile DiD appear separately because fixed-threshold CDF changes do not imply fixed-rank quantile changes. “Nuisance DR” means double robustness of estimation conditional on a base causal restriction; distribution regression is a model for the CDF process and is not, by its name alone, doubly robust.

Table 3: Assumption crosswalk for specialized designs and weaker conclusions

	Family	ID result	Feasible data	Counterfactual restriction	Observable implication/check	Support/shape burden	Still not identified	Fallback
∞	Staggered QDiD	Point within cohort-time cells	Multiple periods/cohorts	No anticipation plus cohort-time CiC, DPT, conditional, or copula-invariance restriction (Section 8.3)	Cohort pretrends, support, comparison sets, weights, transition evidence	Cohort/event-time support	A unique pooled quantile without aggregation rule	Cohort effects; bounds; doubly robust re-estimation
	Extreme CiC	Point under tail model	Baseline CiC design with tail observations	CiC plus tail regularity and threshold choice	Tail plots, effective count, threshold stability	Tail overlap and extrapolation	Robust extreme QTT under arbitrary tail behavior	Interior quantiles; sensitivity range
	Fuzzy CiC	Point or set for compliers	Fuzzy two-by-two design	First stage, monotonicity, and complier rank/distribution restriction	Cell treatment rates, first stage, raw complier-CDF validity	Mixture support; strong enough first stage	Population QTT outside compliers	Bounds or no point estimate
	Weak restrictions only	Set	Any design with usable marginals	Support, shape, or restricted dependence class only	Endpoint validity, width, shared conclusions	Assumption-specific envelope support	Unique counterfactual distribution	Report set; decline point identification

Reading rule: These rows change the design, target, or strength of the conclusion. They are not refinements that can be selected after inspecting which result is most favorable. Cohort aggregation, tail thresholds, complier shares, and dependence classes should be defined before viewing the final QTT curve.

4.2 What a completed analysis will need to show

It helps to know, before working through four sections of identification arguments, what a finished QDiD analysis will eventually have to make visible. A single QTT curve is rarely enough on its own. The reader will need to see which feature of the distribution is policy-relevant – a threshold effect $DTT(y)$, a QTT, a conditional or cohort–time effect, or an identified set – and the population and weighting rule the marginal distribution is formed over, since a clustered policy can be unit-weighted or cluster-weighted with different answers. The observed design then motivates the method rather than the reverse: the number of periods, panel linkage, treatment timing, anticipation rules, and assignment level determine which restrictions are even available, and in staggered designs the comparison cohorts available to each treated cohort should be established before any aggregation is shown. The restriction that constructs the missing distribution should appear in words and as a population operator – the additive CDF construction for DPT, the additive quantile construction for a quantile-trend claim, the rank-transport map for CiC, or the marginal-plus-dependence restriction for a panel method – because a shared nickname can otherwise conceal genuinely different population objects. Evidence should be tied to that specific construction: threshold-specific placebo processes and a raw-CDF shape check for DPT, support and placebo transports for CiC, transition and attrition evidence for panel methods, overlap and weight diagnostics for conditional methods. Estimation and inference belong together – the formula or command, resampling unit, number of draws, and pointwise-or-uniform target reported in one place, with the resampling unit matched to the level of treatment assignment rather than to the unit of observation. And the analysis should end with what survives an alternative assumption: at least one credible competing construction, sensitivity path, or set of bounds, arising from a different substantive story rather than a search for a favorable estimate, so the reader can see which conclusions are robust to the identifying choice and which are not.

Table 4: Verified implementation map (accessed July 21, 2026)

Task	Software/version	Principal command	Built-in inference	Persistent source and limitation
CiC; panel QTT	R <code>qte</code> 1.3.1	<code>CiC()</code> , <code>panel.qtet()</code>	Package bootstrap; user must verify resampling unit	https://CRAN.R-project.org/package=qte ; assignment-cluster wrapper may be required
Additive quantile DiD	R <code>qte</code> 1.3.1	<code>QDiD()</code>	Package bootstrap; user must verify resampling unit	Quantile-change operator, not the threshold-DPT CDF operator
Threshold DPT	V18 replication code	Four empirical CDFs; raw additive CDF operator	State-stratified process bootstrap	Programs/EmpiricalIllustration; shape correction and inversion follow the raw-CDF check
Canonical CiC	Stata SSC <code>cic</code>	<code>ssc install cic</code> ; <code>cic</code>	Command-specific bootstrap	https://ideas.repec.org/c/boc/bocode/s458656.html ; inspect discrete/continuous option
Extreme CiC	Stata SSC <code>ecic</code>	<code>ssc install ecic</code> ; <code>ecic</code>	Command-specific	https://ideas.repec.org/c/boc/bocode/s459194.html ; report tail threshold
Quantile/process tools	R <code>quantreg</code> 6.1; <code>Counterfactual</code> 1.2	<code>rq()</code> , <code>rearrange()</code> , <code>counterfactual()</code>	Method-specific pointwise/process tools	CRAN package pages; generic tools are not QDiD identification routines
Distribution-regression DiD	Author research code/material	Link-scale distribution regression by threshold	Verify current release	https://arxiv.org/abs/2409.02311 ; no survey-created command is implied
Fan–Yu bounds and clustered bands	V18 replication code	bounds-inference script	State-stratified endpoint-process bootstrap	Programs/EmpiricalIllustration; illustration-specific implementation

Table 5 collects this into a compact record. It is deliberately method-neutral – it does not ask every application to estimate every model – and its purpose is to let a reader trace a reported conclusion back through the estimator to the economic counterfactual argument, and see immediately where a different argument or a set-valued conclusion would lead.

Table 5: Minimum QDiD reporting record for readers and referees

Record block	Report explicitly	Question the reader can answer
Estimand	Functional, target population, weighting, quantile grid	What causal or set-valued object is being reported?
Data and assignment	Cell sizes, periods, panel status, cohorts, assignment cluster	Which methods and inference procedures are feasible?
Identification	Counterfactual restriction in words and as an operator	What replaces the missing treated-post untreated distribution?
Diagnostics	Support, placebo process, shape, transitions, overlap, attrition	Which observable implications are contradicted or weakly informed?
Implementation	Command/formula, version, tuning, shape correction, inversion	Does the computation match the named restriction?
Inference	Pointwise/uniform target, resampling unit, draws, seed	What uncertainty is covered, and conditional on which model?
Sensitivity	Competing operator, bounds, or no-point conclusion	Which claims survive uncertainty about identification?

With the naming distinctions fixed, a working decision table in hand, and a target reporting record in view, the next four sections work through each family in the depth that Tables 2 and 3 only summarize. Section 5 develops CiC and its covariate-adjusted extension; Section 6 turns to panel and dependence-based QTT methods; Section 7 covers DPT, distribution-regression DiD, and the covariate-adjusted inverse-probability construction of Kim and Wooldridge (2025); and Section 8 extends the cohort–time framework to staggered adoption, including the copula-invariant and doubly robust constructions of Section 8.3. Section 9 then separates covariate adjustment and double robustness from the specialized extreme- and fuzzy-CiC targets, before the paper turns to software, the empirical illustration, and the Monte Carlo evidence behind the claims made along the way.

5 Rank-Based Changes-in-Changes Method

Rank-based method is a natural starting point for QDiD because they identify the entire counterfactual distribution without reducing the problem to a mean restriction. They are related to, but identified differently from, the broader counterfactual-distribution and decomposition literature (DiNardo et al., 1996; Machado and Mata, 2005; Rothe, 2010; Chernozhukov et al., 2013; Firpo et al., 2011). The economic idea is that untreated outcomes are generated by a stable latent rank mapped into outcomes through a time-varying monotone function. This is especially attractive when the outcome is measured on a scale where monotone transformations are substantively meaningful.

In the absence of treatment, an individual’s latent position in the outcome distribution is mapped into observed outcomes by a time-specific increasing function. The comparison group reveals how that map changes over time. The treated group’s baseline distribution then determines which ranks must be transported through the estimated change. Two groups and two periods are enough for the canonical design, and repeated cross-sections can be used because the procedure transports distributions rather than observed individual trajectories.

Athey and Imbens (2006) identifies the treated-post untreated distribution by transporting the treated baseline distribution through the comparison group’s monotone rank map. In the continuous canonical case,

$$F_{11}^{0,CiC}(y) = F_{10}[Q_{00}\{F_{01}(y)\}].$$

The identifying content is compact but strong: untreated outcomes are generated by a scalar latent rank, the outcome map is strictly increasing, the rank distribution is stable within group, and treated ranks lie in comparison-group support. These assumptions are not mean parallel trends and do not identify the distribution of individual treatment gains.

Inverting this estimated CDF gives counterfactual quantiles, which are subtracted from the observed treated-post quantiles. Equivalently, one can transform each treated baseline observation through the estimated comparison-group quantile map and use the empirical distribution of the transformed observations. This latter representation is often convenient for computation and bootstrap inference.

The simplicity of the plug-in estimator should not obscure where identification enters. The observed comparison transformation is applied to treated ranks only when the relevant ranks lie within comparison-group support and when changes in composition do not alter the latent-rank distribution. With mass points, generalized inverses and possible set identification require care; with sparse tails, a few order statistics can determine the estimated map. Additional pretreatment periods are useful because the same transport can be applied as a placebo and compared with the treated distribution actually observed in the next pretreatment period. A bootstrap should re-estimate every CDF, inverse, and transported outcome rather than treating the estimated quantile map as fixed.

In a sample, the formula can be evaluated directly on an outcome grid or through transported baseline observations. The grid approach makes CDF shape and support visible, whereas the observation-level approach makes it easy to calculate quantiles and other functionals from one simulated counterfactual sample. Ties should be handled with an explicitly stated generalized-inverse convention. Uniform bands are obtained by resampling at the sampling or assignment level, rebuilding the entire transport, and recording the supremum of the studentized CDF or QTT process. Systematic support or placebo-transport failures require a narrower target, an alternative counterfactual model, or bounds rather than a numerical correction.

In the minimum-wage example, the rank argument is that county unemployment ranks in treated states would have passed through the same untreated transformation observed among comparison states. That claim is meaningful only if the comparison transformation spans the treated ranks and if the composition of counties is stable. The method is attractive when such ranks have an economic interpretation and invariance to increasing transformations is desirable; it is not merely a nonlinear version of mean parallel trends.

Covariate-adjusted CiC keeps the rank-transport argument conditional on $X = x$ and aggregates the conditional counterfactual over the treated-post covariate distribution. In schematic form,

$$F_{11}^0(y) = \int F_{11}^0(y | x) dF_{X|G=1,T=1}(x).$$

The extension is useful when untreated rank evolution is credible only after conditioning, but it exchanges an unconditional rank restriction for conditional overlap, specification, and aggregation requirements (Melly and Santangelo, 2015; Firpo, 2007; Cattaneo, 2010). One may estimate the conditional CDFs by distribution regression, local methods, series, or discrete-cell comparisons; within each covariate value, the same quantile transport as above produces $\hat{F}_{11}^0(y | x)$. The final marginal counterfactual averages these conditional distributions using treated-post covariate weights. This last integration step matters: averaging conditional quantiles would generally estimate a different object.

Covariates should therefore be chosen because they make the counterfactual argument more plausible, not simply because a flexible learner can absorb them. Estimation should report overlap in the treated target population, effective sample sizes or weight concentration, monotonicity of estimated conditional distributions, and sensitivity to trimming and smoothing. Cross-fitting can reduce overfitting bias when flexible nuisance estimates are used, but it cannot create conditional support. When support becomes thin, restricting the target or reporting bounds is preferable to extrapolating a high-dimensional conditional rank map.

6 Panel and Dependence-Based QTT Methods

Panel QTT methods move beyond marginal distributional restrictions by using information on transitions through the outcome distribution. The extra panel information is valuable, but it shifts the identifying burden: the researcher must defend restrictions on the dependence between untreated outcome levels and untreated outcome changes. These assumptions are often more interpretable with administrative panels, linked employer-employee data, or longitudinal surveys than with independent repeated cross-sections.

Marginal CDFs reveal how much mass lies below each threshold, but they do not reveal how the same units move through the distribution. A panel supplies this missing transition information. It allows the analyst to study whether high baseline outcomes tend to be followed by large or small untreated changes, whether rank mobility is stable across groups, and whether attrition is concentrated in particular parts of the distribution. This additional information can make a dependence restriction more plausible, although it never makes such a restriction automatic.

For the minimum-wage example, linked county records reveal whether initially high-unemployment counties tend to experience larger subsequent changes. That transition evidence cannot be reconstructed from independent cross-sections. The useful question is therefore not simply whether identifiers are present, but whether pretreatment transitions support the particular dependence structure used to form the missing treated counterfactual.

6.1 Panel QTT estimation

A useful estimation decomposition writes the missing untreated post outcome as $Y_1(0) = Y_0(0) + \Delta Y(0)$. The treated baseline distribution is observed. The distribution of the missing treated-group change may

be recovered from comparison-group changes or from an earlier treated-group transition under a stability restriction. What remains is their dependence. A copula, conditional-rank, or conditional-change restriction couples the estimated change distribution to the observed treated baseline. Computation then draws or matches a counterfactual change for each treated baseline value, adds the two components, and takes the empirical CDF of the resulting counterfactual outcomes.

This decomposition makes the empirical work concrete. Estimate the relevant marginal CDFs and quantile functions, estimate or impose the baseline–change coupling, construct $\hat{Y}_{i1}(0)$ for treated observations, and calculate the counterfactual CDF and its quantiles. If the method uses conditional distributions, this sequence is repeated within covariate or baseline–outcome cells and then integrated over the treated population. Bootstrap inference must preserve complete individual histories and repeat estimation of both the marginals and the dependence component. With treatment assigned at a higher level, entire assignment clusters, rather than individual histories, are resampled.

Panel QTT methods use observed transitions to supplement the marginal distributions. A useful representation writes the untreated post outcome as baseline plus an untreated change,

$$Y_1(0) = Y_0(0) + \Delta Y(0).$$

The treated counterfactual distribution is identified only after specifying how the treated baseline is coupled with its missing untreated change. Callaway and Li (2019) combines stability of the untreated-change distribution with a restriction on the dependence between baseline outcomes and changes. Panel identifiers reveal transitions, but they do not validate either restriction.

The plug-in estimator replaces each marginal distribution and quantile map in the identification formula with its empirical or reweighted counterpart. In the common three-period implementation, the earlier treated transition supplies the treated change ranks, the comparison group supplies the evolution of untreated changes, and a rank or copula restriction maps those changes to the latest treated baseline. The resulting observation-level counterfactuals can be averaged to estimate the ATT or inverted to estimate the QTT process. Supplementary formulas spell out this matching operation in Appendix B.

Pretreatment transition matrices, rank-mobility plots, and baseline–change correlations should accompany the estimates because they show where the coupling is doing work. Attrition must be described as part of the design, not only as a missing-data problem. Sensitivity to alternative copulas is especially informative: if several plausible couplings give different counterfactual tails, the data identify the marginals more strongly than the QTT. In that case, Makarov–Fréchet or related bounds state honestly what is learned without selecting one coupling.

6.2 Two-period panels and the repeated-cross-section extension

The two-period method of Callaway et al. (2018) begins with a genuine panel. For untreated units it observes both the baseline outcome and the subsequent change. Conditional distributional DiD supplies the treated group’s missing change distribution, while conditional copula invariance transfers the observed

baseline-change rank dependence from the comparison group to the treated group. Unlike Callaway and Li (2019), which uses an earlier treated-group transition to stabilize dependence over time, the two-period construction transports dependence across groups. Both are panel arguments, but they use different sources of information.

Repeated cross-sections require an additional step because no individual change ΔY_i is observed. The extension in Callaway et al. (2018) adds conditional rank invariance and replaces the unavailable comparison-group change with a rank-matched pseudo-change. For a baseline comparison observation with covariates x , define

$$\Delta^g Y_i = Q_{01}\{F_{00}(Y_{i0} | x) | x\} - Y_{i0}.$$

The baseline observation's rank in the comparison group is followed from the comparison baseline distribution into the comparison post distribution. That pseudo-change is then combined with the treated baseline outcome at the same conditional rank. The resulting conditional counterfactual is

$$F_{11}^0(y | x) = \Pr[\Delta^g Y_i + Q_{10}\{F_{00}(Y_{i0} | x) | x\} \leq y | G_i = 0, T_i = 0, X_i = x].$$

This formula is estimable from the four repeated-cross-section cells: it uses baseline comparison observations and conditional CDF and quantile maps, not an unobserved individual ΔY_i . Conditional QTTs follow by inversion. An unconditional counterfactual is obtained by averaging $F_{11}^0(y | x)$ over the treated target distribution of X and only then inverting.

The price of dispensing with panel linkage is therefore explicit: fine covariate cells must overlap, the relevant conditional distributions must be estimable, and conditional rank invariance must supply the missing longitudinal coupling. Covariate or cell-probability weights align composition but do not recreate transitions. If this extra rank restriction is not credible, repeated cross-sections more naturally support marginal constructions such as DPT or CiC, or a set-valued conclusion. Appendix B gives the feasible plug-in estimator and keeps it separate from the panel estimator.

7 Distributional Trends and Regression Methods

A second major family keeps the spirit of canonical DiD by imposing restrictions on distribution functions, threshold indicators, or distribution-regression links. It draws on conditional-distribution and counterfactual-distribution methods developed outside DiD (Foresi and Peracchi, 1995; Rothe, 2010; Chernozhukov et al., 2013). These methods are attractive when researchers want continuity with mean parallel trends and when pre-treatment data can be used to examine distributional evolution at many thresholds. The cost is that the restriction is generally not invariant to arbitrary monotone transformations of the outcome.

7.1 Distribution-regression DiD for univariate and multivariate outcomes

Fernández-Val et al. (2025) identifies the counterfactual distribution by fitting a distribution-regression model at each threshold and imposing a no-interaction restriction on its link scale. Writing Λ for the chosen link (the authors use the logistic CDF), the model is

$$F_{Y(0)|G,T,X}(y \mid g, t, x) = \Lambda\{p_\alpha(x)' \alpha(y) + p_\tau(x)' \tau(y) t + p_\gamma(x)' \gamma(y) g\},$$

where $p_\alpha(x)$, $p_\tau(x)$, and $p_\gamma(x)$ are covariate bases for a baseline, a time, and a group component at threshold y . The identifying restriction is that this specification omits a group $\times$ time interaction: absent treatment, the group coefficient $\gamma(y)$ would be the same in both periods at every threshold. That omission is the link-scale analogue of parallel trends, and it is exactly what makes the untreated post-period distribution recoverable from the three observed cells. This makes the identifying scale, link, threshold grid, and covariate support visible in a way that a schematic index cannot.

In practice, the model is fitted repeatedly over a threshold grid. For each y , estimate the link-scale group and time components using the untreated observations and predict the missing treated-post untreated probability for every treated covariate vector. Averaging these predictions yields the marginal counterfactual CDF. Quantiles and their bands are then obtained by inversion of the estimated process, preferably with a bootstrap or influence-function procedure that carries estimation error from all thresholds and nuisance functions through the inversion.

The multivariate extension models joint threshold events and therefore adds a restriction on dependence across outcomes. Estimation may fit joint indicators such as $1\{Y_1 \leq y_1, Y_2 \leq y_2\}$ or combine marginal distribution regressions with a dependence model. Valid marginal trends do not identify a joint counterfactual. This construction is attractive when covariates are important or the policy question is genuinely multivariate, but link and dependence choices are identifying ingredients. Report marginal fit, raw joint-CDF validity, dependence sensitivity, and results across defensible links.

This survey refers to the construction as *DRegDiD* rather than by the authors’ own shorthand “DR-DiD.” The reason is deliberate, not cosmetic: “DR DiD” is already the standard label in the applied DiD literature for the *doubly robust* difference-in-differences estimator of Sant’Anna and Zhao (2020)—distributed, among other places, as the widely used *DRDID* package—which is discussed on its own terms in Section 9.2. Distribution regression and double robustness are unrelated ideas that happen to share three initials, and a survey whose Section ?? is devoted to warning readers away from conflating similarly named objects should not itself create a new instance of the problem. Readers who encounter “DR-DiD” elsewhere should check which of the two constructions the source means before comparing results.

7.2 Covariate-adjusted distributional trends via inverse probability weighting

Kim and Wooldridge (2025) keep the threshold-by-threshold logic of DPT but move the covariate adjustment from a parametric link, as in distribution regression, to inverse probability weighting, and from panel linkage,

as in the methods of Section 6, to repeated cross-sections. Their distributional DiD (DDID) restriction requires that, conditional on $X = x$, the untreated CDF change be the same across groups at every outcome threshold:

$$F_{Y(0)|X=x,1,1}(y) - F_{Y(0)|X=x,1,0}(y) = F_{Y(0)|X=x,0,1}(y) - F_{Y(0)|X=x,0,0}(y), \quad \forall y.$$

This is the same additive threshold restriction as unconditional DPT, imposed inside each covariate cell rather than assumed to hold in the marginal distribution alone; it is compatible with, but strictly stronger than, ordinary mean parallel trends on the conditional mean.

Rather than integrate conditional CDFs cell by cell, as the covariate-adjusted DPT construction above does, Kim and Wooldridge recover the marginal counterfactual by reweighting the same additive operator. Writing $p_{g,t}(x) = \Pr(G = g, T = t \mid X = x)$ for the conditional cell probability and $p_{11} = \Pr(G = 1, T = 1)$,

$$F_{11}^0(y) = E \left[\frac{p_{11}(X)}{p_{11}} \left\{ F_{Y|X,1,0}(y) + F_{Y|X,0,1}(y) - F_{Y|X,0,0}(y) \right\} \right].$$

The weight reweights the covariate-specific DPT operator to the covariate distribution of the treated post-period group, so a single first-stage propensity model does the work that distribution regression assigns to a link function fit anew at every threshold. The construction extends to multi-valued treatments by summing level-specific weighted operators over treatment levels, a coherent way to accommodate heterogeneous treatment intensities without leaving the repeated-cross-section design. Appendix B gives the plug-in IPW estimator.

Because F_{11}^0 is still an additive operator, DDID carries the same testable consequence as unconditional DPT: the implied counterfactual must be non-decreasing and lie in $[0, 1]$. Kim and Wooldridge turn this into an explicit test rather than a post-hoc shape correction, adapting the high-dimensional moment-inequality bounds of Chernozhukov et al. (2017) to a grid that can grow with the sample and comparing the implied CDF at every pair of grid points rather than only adjacent ones. Inference for the CDF, quantile, and QTT processes uses the same generic simultaneous-band recipe as the rest of this survey: bootstrap the estimated CDFs, monotone, invert to quantile functions, and difference—a route that remains valid when outcomes are discrete or mixed, which a quantile-by-quantile normal approximation would not handle cleanly.

The appeal of this construction is what it does not require: no copula or dependence restriction of the kind Section 6 needs, and no parametric link chosen anew at every threshold. It asks only that the additive DPT restriction hold conditionally, and it inherits DPT’s testable shape restriction rather than CiC’s rank-transport diagnostics. It is best read, therefore, as the covariate-adjusted, repeated-cross-section companion to unconditional DPT, not as a competitor to CiC or to the panel constructions above.

7.3 Nonparametric DPT in repeated cross-sections

Distributional parallel trends (DPT) applies the comparison group’s untreated CDF change at every threshold:

$$F_{11}^{0,DPT}(y) = F_{10}(y) + F_{01}(y) - F_{00}(y).$$

The construction is close to canonical mean DiD but stronger because it restricts an entire threshold process. It accommodates discrete and mixed outcomes naturally, after which QTTs are obtained by inverting the estimated counterfactual CDF. Conditional and regression versions replace unconditional threshold evolution with evolution conditional on X or on a chosen link scale (Bonhomme and Sauder, 2011; Fernández-Val et al., 2025); Section 7.2 covers the inverse-probability-weighted covariate adjustment of Kim and Wooldridge (2025) in detail.

The empirical estimator uses sample proportions on a common threshold grid. For covariate-adjusted DPT, one can first reweight comparison observations to the treated covariate distribution or estimate conditional cell CDFs and integrate them over treated covariates. Both approaches should make the target weights visible. With discrete outcomes, inversion uses the first support point at which the estimated CDF reaches the desired probability, so confidence sets are often most naturally constructed for the CDF and then inverted.

The raw operator may leave $[0, 1]$ or violate monotonicity. A simultaneous band for the raw process should therefore precede clipping, cumulative-maximum correction, isotonic regression, or rearrangement. Small sampling defects can be regularized; systematic shape failure contradicts an observable implication of the maintained model. Report threshold-specific pretreatment trends, overlap, weight concentration, the raw and regularized counterfactual CDFs, and uniform rather than selected-quantile inference.

8 Staggered-Adoption and Dynamic QDiD

Staggered adoption changes the object of interest: the primitive is not one distributional treatment effect but a collection of cohort-time counterfactual distributions. This mirrors the modern mean-DiD literature on cohort-time effects, heterogeneous timing, and transparent aggregation (Callaway and Sant’Anna, 2021; Goodman-Bacon, 2021; Sun and Abraham, 2021; de Chaisemartin and D’Haultfoeuille, 2020; Borusyak et al., 2024): define cohort-time effects first, then aggregate transparently, and the discipline matters even more here because heterogeneity can differ across both event time and quantile.

The reason is not tidiness. Goodman-Bacon (2021) and de Chaisemartin and D’Haultfoeuille (2020) showed that pooling staggered adoption into one two-way fixed-effects regression can assign *negative* weight to some cohort-time comparisons, so that a treatment effect positive for every cohort at every date can still average to a negative pooled coefficient. A pooled quantile regression inherits the same contamination at each fixed τ : it is still a weighted combination of the same forbidden already-treated-versus-already-treated comparisons, now applied to a conditional quantile rather than a conditional mean. The cohort-time construction below is the distributional analogue of the fix, not a bookkeeping preference – it builds every

comparison from a valid not-yet-treated or never-treated baseline before any aggregation step touches the estimates.

8.1 Cohort-time distributions as the identification primitive

Once treatment begins at different dates, there is no single treated-post cell. Each adoption cohort g and calendar period t has its own observed distribution and, once adopted, its own missing untreated counterfactual $F_{g,t}^0$. Estimation proceeds cell by cell: cohort g is compared at time t with units untreated at t , using a baseline period before g adopts, and a stated distributional operator – DPT, CiC, or a conditional alternative – turns that comparison into $\hat{F}_{g,t}^0$. The cohort-time effect is

$$QTT_{g,t}(\tau) = Q_{g,t}^1(\tau) - Q_{g,t}^0(\tau), \quad t \geq g,$$

identified under no anticipation, a valid not-yet-treated or never-treated comparison set, cohort-specific support, and the chosen distributional restriction, following the design discipline of modern mean DiD (Callaway and Sant’Anna, 2021; Goodman-Bacon, 2021; Sun and Abraham, 2021; de Chaisemartin and D’Haultfoeuille, 2020; Borusyak et al., 2024). The natural plug-in estimator repeats the selected two-period estimator for every admissible (g, t) pair; comparison-group membership can change with calendar time under not-yet-treated designs, so estimated comparison and covariate weights should be saved with the results rather than treated as background detail. Cohort-specific support, comparison sets, pretrends, and the disaggregated distributions belong in the reported output alongside any summary – in a staggered minimum-wage setting, for instance, showing the cohort-time distributions before aggregation lets a reader see whether a smooth event-time curve combines genuinely similar effects or averages offsetting ones.

8.2 Aggregating across cohorts and event time

Cohort-time DTTs can be averaged at a common threshold because they are linear differences in probabilities. Quantiles are not: averaging $QTT_{g,t}(\tau)$ across cohorts answers a different question from mixing the cohort-time observed and counterfactual distributions and taking the quantiles of the mixtures. The former, an event-time QTT, summarizes cohort-specific quantile contrasts weighted to represent cohort size or another policy population; the latter, a mixture-based effect, compares quantiles for a synthetic mixture population. Both can be useful, but the paper must state which one it reports and the weights behind it. Inference should resample the underlying observational or assignment clusters, reconstruct every cohort-time counterfactual, and repeat the full aggregation, so that reuse of the same cohorts across event-time cells is reflected in the joint band. The main threats are sparse cohorts, changing support, anticipation, and opaque aggregation; a credible application shows cohort-time CDFs or QTTs before pooling and treats sign reversals across cohorts as heterogeneity to report, not noise to average away.

8.3 Concrete constructions: copula invariance and doubly robust cohort-time QTT

The constructions above state what a cohort-time QDiD estimator must do without fixing one. Three recent papers fix one, and each addresses a gap the last leaves open.

Li and Lin (2024) identify $F_{g,t}^0$ by extending the two-period, covariate-invariant construction of Callaway et al. (2018) – not the copula-*stability* construction of Callaway and Li (2019) – to staggered timing. Writing $\Delta^{(g-1,t)}Y(0) = Y_t(0) - Y_{g-1}(0)$ for the long difference between the period just before cohort g adopts and calendar period t , their restriction pairs conditional distributional parallel trends against a not-yet-treated group (eligible at least through some period $s \geq t$) with a conditional group-invariant copula linking that long difference to the pre-adoption level $Y_{g-1}(0)$. Sklar’s theorem then lets the comparison group’s observable dependence substitute for cohort g ’s unobservable one – the same move Section 6 uses to extend two-period CiC-style identification from copula stability within one group over time to invariance across groups. The more consequential result may be the negative one: generalizing Callaway and Li (2019)’s copula-*stability* construction to staggered timing the same way does *not* work, because quantiles lack the linearity needed to chain single-period-apart copulas into a multi-period-apart one. Li and Lin’s contribution, published in *Economics Letters*, is an identification result; it stops short of an estimator or an inference procedure.

Ciaccio (2025) supplies both. Working with a never-treated comparison group (the construction extends symmetrically to not-yet-treated units), Ciaccio adds a limited-anticipation parameter $\rho \geq 0$ to the same restriction, then gives a sample-analogue plug-in estimator for $F_{g,t}^0$ and a Callaway–Li-style studentized bootstrap band for $QTT_{g,t}(\tau)$ (Appendix B), together with explicit exposure-length and overall-weighted aggregation formulas that make the previous subsection’s aggregation step concrete. Because rank invariance between a cohort’s treated and counterfactual outcomes is often not credible, Ciaccio also proposes a rank-free fallback: first- and second-order stochastic dominance tests between $F_{Y_t(g)|G=g}$ and $F_{g,t}^0$, building on Maasoumi and Wang (2019), that ask whether one distribution lies uniformly above another rather than which quantile moved by how much.

Galvao and Miller (2026) take the construction in a different direction: rather than a single propensity or conditional-CDF model, they estimate $F_{\Delta^{(g-1,t)}Y(0)|G=g}$ with a doubly robust moment condition, consistent when either a generalized propensity score or a conditional-CDF model for the long difference is correctly specified. This is the same augmented-inverse-probability-weighting construction Section 9.2 describes for two-period mean DiD, generalized from a conditional mean to every threshold and from a single date to staggered cohorts, using Callaway and Sant’Anna (2021)’s $ATT(g, t)$ aggregation weights. It is a legitimate use of “doubly robust” in the Sant’Anna–Zhao sense – unlike the distribution-regression construction of Section 7.1, which this survey deliberately does not call DR-DiD – but it does not relax the identifying restriction: the source paper’s Assumptions IDS.3–IDS.5 are the same distributional-parallel-trends-and-copula-invariance pair used above, so nuisance-model robustness is still not restriction robustness. One practical subtlety worth flagging: under the empirical bootstrap, the usual double-robustness relaxation of the required nuisance convergence rate does not carry over to valid standard errors, so both nuisance

components are still held to a uniform $o_p(n^{-1/4})$ rate.

Read together, the three papers separate by contribution rather than by restriction: Li and Lin show the restriction identifies $F_{g,t}^0$ under staggered timing and why copula stability alone does not generalize; Ciaccio turns it into an estimator, an aggregation scheme, and a rank-free alternative; Galvao and Miller trade a heavier nuisance-estimation burden for protection against misspecifying either nuisance model. None weakens the underlying distributional-parallel-trends-and-copula restriction, so the diagnostics recommended for panel QTT methods in Section 6 – pretreatment transition evidence and sensitivity to the assumed dependence structure – remain the relevant checks here. Li and Lin (2024) is peer reviewed; Ciaccio (2025) and Galvao and Miller (2026) are working papers, treated in this survey as evolving contributions.

9 Covariate Adjustment and Specialized Targets

This section separates two ideas that are easily confused. Covariate adjustment and doubly robust scores concern how a counterfactual distribution is estimated after a causal restriction has been chosen. Extreme-value and fuzzy-treatment methods add identifying structure and may change the target itself. Calling all of them “robust” would obscure the difference between protection against nuisance-model misspecification and robustness of the causal conclusion.

9.1 Why estimation robustness is not design robustness

The methods in this section solve different practical problems and should not be read as competing identification strategies. Doubly robust estimation addresses how conditional CDFs and treatment or group probabilities are estimated. Extreme-value methods address what can be learned beyond densely observed quantiles. Fuzzy methods address the fact that assignment and treatment receipt differ. Each begins with a baseline counterfactual restriction and then adds structure tailored to the complication at hand.

This ordering matters empirically. An orthogonal score can reduce sensitivity to nuisance-model error but cannot repair an invalid distributional trend. A Pareto approximation can stabilize an extreme quantile but cannot create tail comparability between groups. A compliance decomposition can recover a complier distribution only under a valid first stage and assumptions linking observed mixtures to latent compliance types. Estimation should therefore be described in two layers: first the causal restriction that constructs the relevant counterfactual distribution, and then the statistical device used to estimate it.

For the minimum-wage example, these extensions become relevant when covariate adjustment is essential, when the policy question concerns an extreme tail with few observed counties or workers, or when exposure is incomplete. The additional assumptions should be visible next to the resulting estimand because the target may change from all treated units to compliers, or from an empirical quantile to a tail extrapolation.

9.2 Covariate adjustment and doubly robust distributional estimation

Doubly robust procedures combine a model for treatment-group or cell probabilities with a model for the relevant conditional distribution. The counterfactual CDF remains consistently estimated when either nuisance component is correctly specified, subject to the maintained causal distributional-trend restriction (Sant’Anna and Zhao, 2020). Quantiles follow by inversion. This is the “DR” that this survey means whenever it writes “doubly robust”; Section 7.1 explains why the unrelated distribution-regression construction of Fernández-Val et al. (2025) is deliberately *not* abbreviated the same way.

At each threshold y , a doubly robust estimator combines an outcome-regression estimate of a conditional CDF with inverse-probability weights derived from a propensity or group-membership model. The augmentation term corrects the regression estimate using weighted residuals $1\{Y \leq y\} - \hat{F}(y | X)$. This produces an estimated counterfactual CDF that remains consistent when either the probability model or the conditional-distribution model is correctly specified, provided the underlying causal distributional restriction is valid. Repeating the score over a threshold grid yields a CDF process; shape correction and inversion then produce QTTs.

Cross-fitting is useful when the nuisance functions are estimated by flexible learners because it separates score evaluation from nuisance fitting, in the same spirit as the double/debiased machine-learning framework for treatment-effect estimation (Chernozhukov et al., 2018): the orthogonal score removes the first-order sensitivity of the estimated CDF to nuisance-model error, which is what licenses flexible learners for the propensity or conditional-CDF components in the first place. Standard errors should be based on the influence function or a bootstrap that repeats sample splitting, nuisance estimation, CDF construction, and inversion. Double robustness concerns nuisance estimation, not the causal design: it cannot repair an invalid DPT, rank, or dependence restriction. Report overlap, weight tails, both nuisance specifications, cross-fitting or regularization choices, and sensitivity when both models may be wrong.

9.3 Specialized target: extreme changes-in-changes

Extreme CiC combines the baseline rank-transport design with tail regularity and extrapolation beyond well-populated empirical quantiles (Sasaki and Wang, 2024). The additional assumptions concern tail family, threshold k , and stability of the tail transformation. These are consequential identifying and tuning choices, not numerical details.

Estimation first applies empirical CiC over the well-supported interior. Above a high threshold, the relevant tail distribution or tail quantile function is replaced by an extreme-value approximation whose parameters are estimated from the largest k observations. The comparison-group tail transformation is then transported to the treated baseline tail, after which the counterfactual extreme quantile is extrapolated and compared with the treated-post extreme quantile. Uncertainty must reflect both the estimated rank transport and tail parameters.

Report effective tail counts, Hill or log–log diagnostics where appropriate, estimates over several defensible thresholds, and interior empirical-CiC results. The threshold k determines the bias–variance tradeoff: using

too few observations creates instability, while using too many makes the tail approximation less credible. If conclusions change sharply with k or tail family, the useful output is a sensitivity range rather than a single extreme QTT.

9.4 Specialized target: fuzzy changes-in-changes

Fuzzy CiC changes both the target population and the identifying burden. Under a valid first stage, monotonicity, and the relevant rank or distributional restrictions, observed mixtures can identify complier counterfactual CDFs or informative bounds. The published anchor for this identifying logic is de Chaisemartin and D’Haultfoeuille (2018), which develops the compliance-mixture argument for fuzzy DiD designs and has an accompanying Stata implementation (de Chaisemartin et al., 2019); the rank-based construction specific to CiC is worked out in the companion manuscript de Chaisemartin and D’Haultfoeuille (2014), which remains unpublished and is treated here as an evolving contribution rather than a settled result. The target throughout is a complier distributional effect, not the marginal QTT for all treated units.

Estimation begins by using treatment rates in the group–time cells to recover the shares of latent compliance types under monotonicity. Observed outcome CDFs are mixtures of these types. Subtracting the estimated always-taker and never-taker components and dividing by the estimated complier share yields complier CDFs or their bounds. A CiC-style rank map can then construct the missing complier counterfactual distribution. Because the denominator is the first stage, small complier shares amplify both sampling noise and any misspecification, and the raw mixture estimate may leave $[0, 1]$ or cease to be monotone.

Applied work should report treatment rates in all cells, the implied complier share, raw complier-CDF range and monotonicity, and sampling uncertainty that reflects a potentially weak first stage. The bootstrap must re-estimate compliance shares and all mixture components before inversion. “Fuzzy DiD” denotes the broader compliance problem; “fuzzy CiC” denotes the rank-based construction. When the first stage is weak or the mixture produces an invalid CDF, bounds or no point estimate are preferable.

10 Software, Diagnostics, and Reporting

A practitioner-oriented application should show the observed treated distribution, estimated counterfactual distribution, QTT or DTT process over a prespecified support, overlap evidence, design-specific diagnostics, and uncertainty bands. For staggered adoption, cohort–time distributions should precede aggregation.

The table distinguishes executable implementations of named estimators from component tools. The `qte`, `cic`, and `ecic` entries implement published method families; the archived scripts implement the stated empirical DPT and bound constructions; `quantreg` and `Counterfactual` provide regression, inversion, and process-inference components but do not by themselves identify a QDiD counterfactual. Author research code is listed only when its link-scale identifying model is named explicitly. Software version numbers and repository links are perishable in a way that identifying assumptions are not: this table is a snapshot as of the stated access date, and a reader consulting it later should re-verify the command, version, and default

resampling unit against the package’s current documentation rather than trusting the table indefinitely. The numerical results in Sections 11–12 and their replication archive, by contrast, are carried forward unchanged from the V18 release; this revision changes exposition, framing, and citations, not the underlying computation, and every reported estimate can still be traced to the same seeded scripts.

Minimal implementation skeleton. Package commands implement a maintained statistical model; they do not choose the identifying assumption or assignment unit. A reproducible R workflow begins as follows:

```
library(qte)                                # version 1.3.1 in the archive
p <- seq(.10, .90, .10)
fit_cic <- CiC(y ~ treated, t = 2007, tmin1 = 2006,
              tname = "year", data = d, panel = TRUE,
              idname = "id", probs = p, se = FALSE)
fit_panel <- panel.qtet(y ~ treated, t = 2007, tmin1 = 2006,
                      tmin2 = 2005, tname = "year", data = d,
                      idname = "id", probs = p, se = FALSE)
# Wrap the full estimator in an assignment-cluster bootstrap when policy
# varies above the observation level; do not accept a unit bootstrap default.
```

For DPT, estimate the four CDF cells on a stated threshold grid, form the raw operator $\hat{F}_{10} + \hat{F}_{01} - \hat{F}_{00}$, construct its simultaneous band, and only then apply a stated shape correction and invert. The V18 archive supplies the executable DPT implementation, its raw-CDF diagnostic, the assignment-cluster process bootstrap, and the bound-endpoint bootstrap rather than a pseudo-command. The separate listing of `QDiD()` prevents the command’s name from being mistaken for the DPT operator.

Table 6: Restriction-specific diagnostics and decisions

Restriction	Operational diagnostic	Implementation	If diagnostic fails
Support/overlap	Effective observations and common outcome/covariate support	Plot cell CDFs; tabulate trimming and propensity tails	Restrict target/support or report bounds
DPT	Pretreatment threshold differences and implied-CDF shape	CDF event studies on a prespecified grid; check range and monotonicity	Qualify DPT, change comparison group, or use sensitivity/bounds
CiC	Stability of control quantile maps and treated composition	Placebo transports; support and composition plots	Observable rank-transport implication is contradicted
Panel dependence	Stability of baseline–change association and attrition	Pretreatment Kendall/Spearman statistics; transition plots by group	Do not treat panel availability as validation
Staggered adoption	Cohort support, pretrends, and weights	Cohort–threshold event studies and weight table	Avoid opaque aggregation; redefine comparison set
Extreme tails	Tail-index and threshold stability	Hill/log–log plots across several k ; effective tail count	Report interior estimates or a wide sensitivity range
Fuzzy design	First stage, monotonicity, and complier-CDF validity	Cell treatment rates, complier share, CDF range/shape, bounds	Weak or invalid mixture calls for bounds or no QTT

Diagnostics do not prove an identifying assumption: they reveal observable implications and identify where extrapolation is doing the work. They should be tied to a prespecified decision. For example, an invalid DPT CDF is not silently corrected: rearrangement is numerical regularization only when the departure is consistent with sampling noise; a systematic violation calls for a different model or an identified set. Likewise, a large placebo transport error should narrow the support or weaken the causal claim.

Illustrative simultaneous diagnostic. This is a pedagogical implementation of established empirical-process and multiplier-bootstrap methods, not a new test (Chernozhukov et al., 2017, 2020, 2019). The caution against treating pretrend non-rejection as design validation also parallels the mean-DiD evidence in Roth (2022). With two pretreatment transitions, define the DPT placebo process

$$\hat{\Delta}_{pre}(y) = \{\hat{F}_{1,-1}(y) - \hat{F}_{1,-2}(y)\} - \{\hat{F}_{0,-1}(y) - \hat{F}_{0,-2}(y)\}, \quad y \in \mathcal{Y}.$$

The omnibus statistic is $T_{pre} = \sup_{y \in \mathcal{Y}} |\hat{\Delta}_{pre}(y)/\hat{\sigma}(y)|$, with a multiplier or cluster bootstrap critical value that controls rejection over the full prespecified grid. CiC placebo transports can be evaluated by replacing $\hat{\Delta}_{pre}$ with the distance between the transported and observed treated-group CDFs. The protocol is: pre-specify $\mathcal{Y}$, size α , and a substantively meaningful tolerance δ ; reject the restriction when the simultaneous test rejects; and flag it as substantively unstable when the confidence band permits deviations larger than δ . Failure to reject is only a screen—post-period violations can occur even when pretreatment restrictions hold—so the preferred estimate must still be accompanied by a sensitivity region or bounds.

Estimated-CDF shape has an analogous rule. Construct a simultaneous band for the raw counterfactual process before imposing monotonicity. Rearrangement or projection is a numerical regularization only when a valid CDF lies inside that band (Chernozhukov et al., 2010). If the band itself excludes every non-decreasing function in $[0, 1]$, the maintained model is rejected on the reported support; clipping the estimate is not an admissible repair. Section 12 reports finite-sample size and power for the multiplier diagnostic and the frequency with which the DPT operator fails this shape requirement.

Assignment-level inference. Resampling must follow the level at which treatment and common shocks vary. When a state, firm, school, or village is the assignment unit, resampling individual observations treats correlated outcomes as independent and generally understates uncertainty. The default is a cluster bootstrap, multiplier bootstrap with cluster scores, or a small-cluster procedure justified for the design. The application should report the number of treated and comparison clusters, whether resampling is stratified by treatment group, and a leave-one-cluster-out sensitivity analysis when influential clusters are plausible.

Pointwise and uniform inference. A pointwise interval covers $\theta(u)$ for one fixed u ; it does not cover an inspected quantile curve. Uniform inference for counterfactual distributions, quantile processes, and discrete outcomes is developed in Chernozhukov et al. (2013, 2020); Donald and Hsu (2014). A simultaneous band over $\mathcal{U}$ uses a critical value for a process such as

$$T_n = \sup_{u \in \mathcal{U}} \left| \frac{\sqrt{n}\{\hat{\theta}(u) - \theta(u)\}}{\hat{\sigma}(u)} \right|,$$

usually approximated by a bootstrap that respects the sampling and treatment-assignment clusters. Researchers should name the index set, resampling unit, number of draws, and whether the reported interval is pointwise or simultaneous. For discrete outcomes or non-smooth inversion, construct a band for the coun-

terfactual CDF first and then invert or project it; a quantile-by-quantile normal approximation can conceal mass-point uncertainty.

Reproducibility standard. An archive should record the software and package versions, access date for external code, random seeds, raw-to-analysis-data script, estimator call, tuning parameters, and machine-readable estimates behind each table and figure. The illustration below uses R 4.3.2 and `qte` 1.3.1 (accessed July 2026); its scripts and CSV outputs are included with the draft. These fields matter because similarly named QDiD routines can encode different panels, counterfactual restrictions, or bootstrap defaults.

11 Empirical Illustration: Minimum Wages and County Unemployment

This section applies the paper’s decision sequence to the replication data of Callaway and Li (2019). It is an illustration of how method choice should be reported, not a new evaluation of minimum-wage policy. The outcome is the February county unemployment rate from the BLS Local Area Unemployment Statistics; county characteristics come from the 2000 County Data Book; and state minimum-wage timing comes from Dube et al. (2016). The balanced 2005–2007 panel contains 2,360 counties: 678 counties in 11 states whose minimum wage first increased in the first quarter of 2007 and 1,682 counties in 20 comparison states. The longer 2000–2007 panel is used only for pretreatment diagnostics.

Design choices and target population. The observed transitions make panel QTT a feasible starting point, using 2005 and 2006 to construct the 2007 untreated counterfactual. Canonical CiC and threshold DPT are estimated on the same counties and quantile grid because they replace the missing distribution using distinct restrictions. A covariate-adjusted panel specification uses region, log 2000 population, log 1997 median income, polynomial terms, and interactions; it is reported as a within-method adjustment check, not as a controlled comparison with the unadjusted CiC and DPT curves. The Fan and Yu (2012) bounds maintain equality in the distribution of untreated changes but leave the dependence between the treated baseline and its counterfactual change unrestricted. The target is the marginal QTT for a county drawn from the treated-state sample. It is county-weighted—states with more observed counties receive more weight—and is neither a state-average policy effect nor the distribution of county-specific treatment effects.

Small-cluster warning: methodological illustration only. Treatment timing varies at the state level, leaving only 11 treated and 20 comparison clusters. Cluster asymptotics and bootstrap refinements can be unreliable with few or highly unbalanced treated clusters (Cameron et al., 2008; MacKinnon and Webb, 2018). State-specific macroeconomic shocks, anticipation of a scheduled minimum-wage change, cross-border adjustment, or other spillovers would invalidate the simple no-interference and counterfactual-trend stories. The replication data do not resolve these threats. The calibrated simulation in Section 12 obtains 0.94

Table 7: How the empirical estimators use the common county sample

Method	Observed inputs	What constructs the 2007 counterfactual	Target and weighting
Panel QTT	Linked county histories, 2005–2007	Earlier treated transition, comparison changes, and baseline–change dependence restriction	Marginal QTT; treated counties receive equal unit weight
Panel QTT with X	Same histories plus region, population, income, and interactions	Conditional/reweighted version of the panel construction	Same marginal target after integrating over treated-county X
CiC	Treated and comparison county distributions, 2006–2007	Comparison-group rank map applied to the treated 2006 distribution	Same county-weighted treated population
Threshold DPT	Four county CDF cells, 2006–2007	Comparison CDF change added at each fixed unemployment threshold	Same county-weighted treated population
Fan–Yu bounds	Treated 2006 levels and comparison 2006–2007 changes	Makarov–Fréchet couplings leave baseline–change dependence unrestricted	Identified set for the same marginal QTT

uniform coverage (MCSE 0.024) for the assignment-cluster bootstrap under its simplified valid-DPT DGP, but that result is not a guarantee for this application. The bands are therefore illustrative assignment-level uncertainty summaries, supplemented by leave-one-state-out influence ranges, rather than a definitive small-cluster validation. The exercise shows how conclusions change across maintained restrictions and when the researcher should decline point identification; it does not establish a preferred minimum-wage estimate.

Table 8: Selected QTT estimates (differences of marginal quantiles) and dependence-robust bounds

	$\tau = 0.10$	$\tau = 0.50$	$\tau = 0.90$
Panel QTT	−0.437	0.307	0.358
Panel QTT with covariates	−0.450	0.317	0.270
Changes-in-changes	−0.188	0.333	0.428
DPT	−0.001	0.201	0.250
Estimated lower endpoint	−1.516	−1.008	−1.803
95% band for estimated lower endpoint	[−2.355, −0.677]	[−1.627, −0.389]	[−2.487, −1.118]
Estimated upper endpoint	1.267	1.409	1.940
95% band for estimated upper endpoint	[0.428, 2.105]	[0.897, 1.922]	[1.182, 2.698]

Notes: Entries are effects in percentage points of the county unemployment rate. Point estimators use the same treated counties, comparison counties, and quantile grid; they differ in the counterfactual restriction. The two estimated endpoints delimit the sample analogue of the identified set; the following rows are confidence bands for those estimated endpoint processes, not additional identified sets. The endpoint bands use 999 state-stratified cluster-bootstrap draws and a single studentized supremum critical value over both endpoints and all nine quantiles. The full grid and machine-readable output are in the replication archive.

Panel QTT and CiC produce negative lower-tail contrasts, whereas the threshold-DPT estimate at the 10th percentile is essentially zero. All point estimators are positive at the median and upper tail, but they disagree materially on magnitude. At the 10th percentile, the estimates range from −0.450 to −0.001 percentage points; at the median they range from 0.201 to 0.333. Covariate adjustment changes the panel curve little over most of the grid. This cross-method agreement is descriptive evidence, not identification: the bounds include zero and economically different effects at every reported quantile. Assignment-level uncertainty about the bound endpoints does not change that decision. For example, the 95% uniform bands for the lower and upper endpoints are [−2.355, −0.677] and [0.428, 2.105] at $\tau = 0.10$, and [−1.627, −0.389] and [0.897, 1.922] at the median. Thus the data do not select a unique causal distribution without an additional rank, trend, or dependence restriction. Because one supremum critical value controls both endpoint pro-

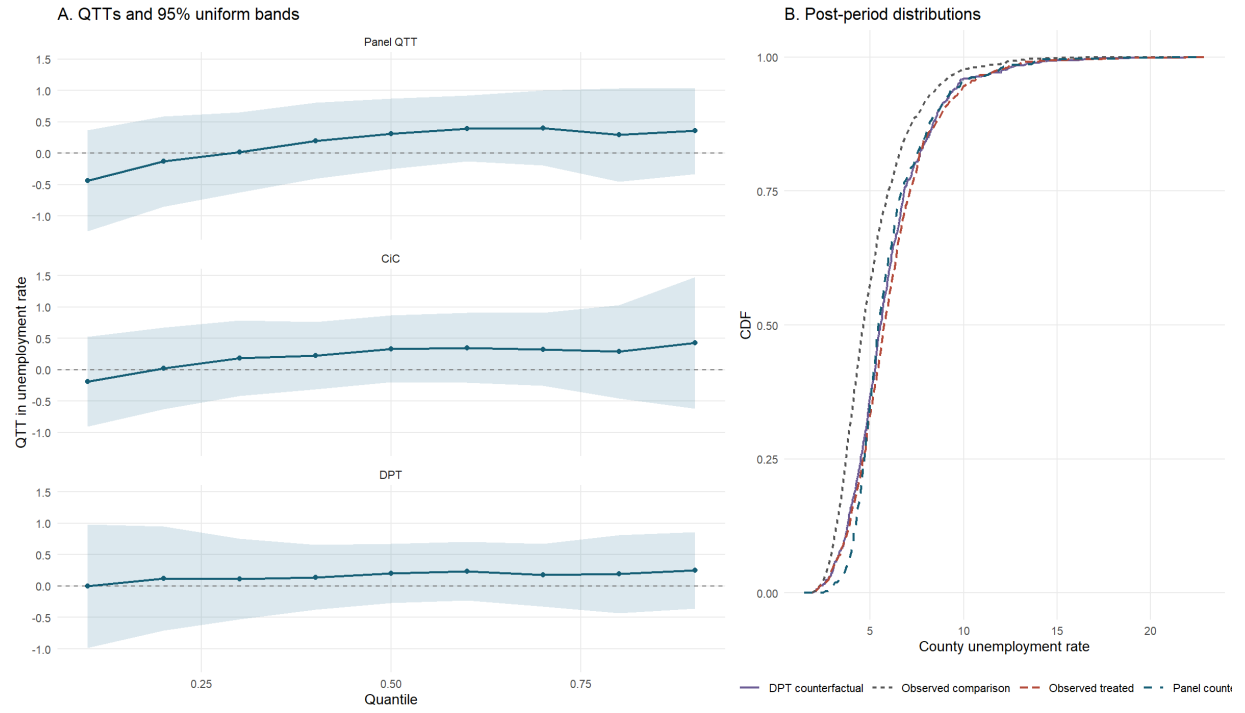


Figure 1: Minimum-wage illustration with assignment-level uncertainty. Panel A reports point estimates and 95% state-clustered uniform bands over $\tau \in \{0.1, \dots, 0.9\}$ separately for each counterfactual restriction. Panel B keeps the observed treated, observed comparison, panel counterfactual, and shape-corrected DPT counterfactual CDFs visible.

cesses over all nine quantiles, taking the lower limit of the lower-endpoint band and the upper limit of the upper-endpoint band gives a simultaneous outer confidence region for the identified QTT set, conditional on the Fan–Yu restrictions and the state-bootstrap approximation. It is not a confidence interval for the distribution of individual county effects. Leave-one-state-out estimates show that no single state explains the broad pattern, but the exact magnitudes remain influential. At $\tau = 0.10$, the ranges are $[-0.620, -0.271]$ for panel QTT, $[-0.307, 0.016]$ for CiC, and $[-0.141, 0.240]$ for DPT. At the median the corresponding ranges are $[0.180, 0.434]$, $[0.193, 0.421]$, and $[0.084, 0.309]$. These ranges are influence diagnostics, not confidence intervals.

Raw DPT operator and shape check. The DPT estimate is formed from the four empirical CDFs, not from differences in marginal quantiles. On the full outcome grid, the raw operator ranges from -0.0006 to 1.0000 and has many numerically small downward steps; the largest is -0.0012 . These departures are reported before shape correction. The diagnostic grid consists of 99 empirical outcome quantiles selected once from the original sample and then held fixed throughout the 999-draw state-stratified bootstrap. The uniform band contains at least one valid nondecreasing CDF in $[0, 1]$. The point counterfactual is therefore clipped and corrected using the cumulative maximum before inversion, while every bootstrap draw repeats the full construction. This is numerical regularization supported by the raw-process band, not evidence that DPT is causally valid.

Pretreatment and transition diagnostics. Figure 2 asks whether the restrictions have observable support. Point placebo QTTs for terminal years 2002–2006 vary across years and are largest in some tail comparisons (for example, the 2002 upper-tail CiC placebo is -0.786). With only 31 assignment clusters, however, the 499-draw method-specific simultaneous bands cover zero for every displayed placebo process. This is a failure to reject, not evidence that any restriction is true: the bands also include economically consequential departures. The transition statistic changes in the eventually treated group: Kendall’s correlation between baseline unemployment and the subsequent change moves from -0.282 in 2006 to 0.027 in 2007, while the comparison-group statistic moves from -0.220 to -0.165 . The post-policy shift is not itself a test of the untreated copula. Taken together, the diagnostics are too weak to treat the panel restriction as automatic, but they do not statistically single out and reject one method either.

Inference and decision. Because policy timing varies at the state level, uncertainty is computed with 999 draws that resample states separately within treated and comparison groups and re-estimate the full quantile process. Simultaneous critical values use the bootstrap supremum statistic over $\tau \in \{0.1, \dots, 0.9\}$; county-level resampling would overstate the effective sample size. The 95% uniform bands include zero at every quantile for all three point estimators. At the 10th percentile they are $[-1.240, 0.367]$ for panel QTT, $[-0.902, 0.526]$ for CiC, and $[-0.984, 0.982]$ for DPT; at the median they are $[-0.253, 0.868]$, $[-0.202, 0.869]$, and $[-0.272, 0.675]$, respectively. These bands address sampling uncertainty conditional on each identifying model; they do not average across models or validate any restriction. The inferential target, state cluster, grid,

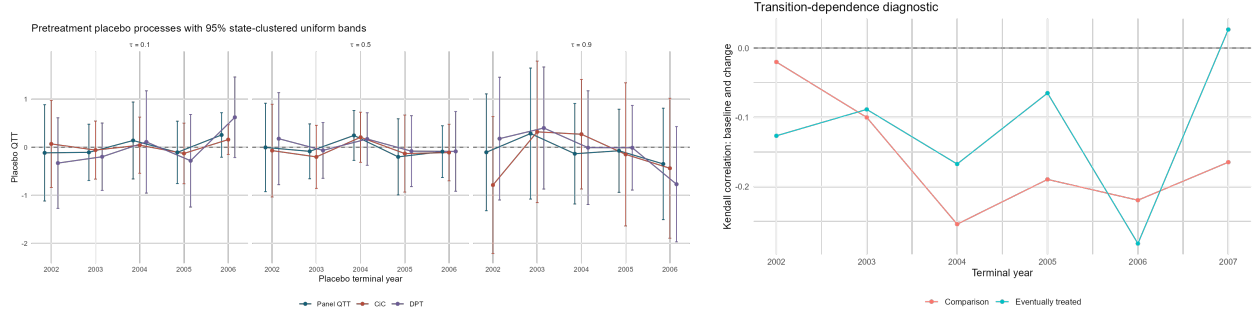


Figure 2: Minimum-wage illustration: pretreatment placebo QTTs (differences of marginal quantiles) with 95% method-specific state-clustered uniform bands (left) and transition-dependence evidence (right). Diagnostics are reported as evidence about the maintained restrictions, not as formal proof that one method is correct.

draw count, and seed are recorded in the replication script. The appropriate conclusion is conditional: panel QTT and CiC have similar lower-tail signs while the DPT estimate is near zero, but sampling uncertainty, wide bounds, and imperfect placebo evidence make the exact shape assumption-sensitive. A reader who accepts the transition restriction may use panel QTT; a reader who accepts threshold parallel evolution may use the DPT curve; a reader who accepts neither should retain the bounds or decline point identification. This is precisely the decision the paper’s taxonomy is designed to expose.

Reproducibility. The application is reproduced by the point-estimation and state-cluster-bootstrap R scripts in the empirical-illustration replication directory. The DPT scripts carried forward in the V18 archive construct the threshold-DPT CDF, its uniform raw-process band, the inverted QTT process, placebo estimates, and leave-one-state-out results. The archive stores the input data, estimator calls, random seed, CSV estimates, bootstrap objects, and figure files. No manual values are used to generate the displayed figures.

12 Monte Carlo Evidence: Four Lessons for Method Choice

The simulation is an assumption laboratory, not a universal horse race, and it is worth being explicit about what that distinction buys and what it does not. Every design below is built by hand so that a specific assumption is true or false by construction; none of the four lessons that follow would surprise a reader who already trusts the identification arguments in Sections ??–10. What the simulation adds is not a new theoretical fact but a concrete answer to “how much does it matter, and how fast does it get better or worse”—questions identification theory alone does not answer. The reported bias, RMSE, and coverage figures are therefore precise about *these* designs, to support exact replication, and are not evidence that generalizes to designs the reader has not simulated. Direct comparisons use the same observables and target the same difference of marginal quantiles; specialized modules reproduce the extra requirements of panel, staggered, tail, fuzzy, and multivariate designs. Exact data-generating processes, estimator formulas, packages, seeds, replication counts, and machine-readable results are in Appendix C and the replication

archive. Naive quantile DiD is defined here as the treated-group change in marginal quantiles minus the comparison-group change at the same quantile; it is a descriptive benchmark, not DPT estimated through a CDF process.

12.1 Lesson 1: Each method works under its own restriction

The core experiment makes both the CiC rank restriction and DPT valid, only CiC valid, only DPT valid, or neither valid. At 4,000 observations per cell, mean absolute bias across five quantiles is 0.001 for CiC and 0.122 for DPT when only rank transport holds. When only DPT holds, the ordering reverses: 0.311 for CiC and 0.002 for DPT. When neither holds, the biases are 0.269 and 0.376. Under both restrictions, all three estimators have mean absolute bias at most 0.002 and mean RMSE between 0.051 and 0.056; the largest Monte Carlo standard error for bias or RMSE in these displayed designs is 0.005. Figure 3 therefore illustrates method–assumption matching, not an unconditional ranking.

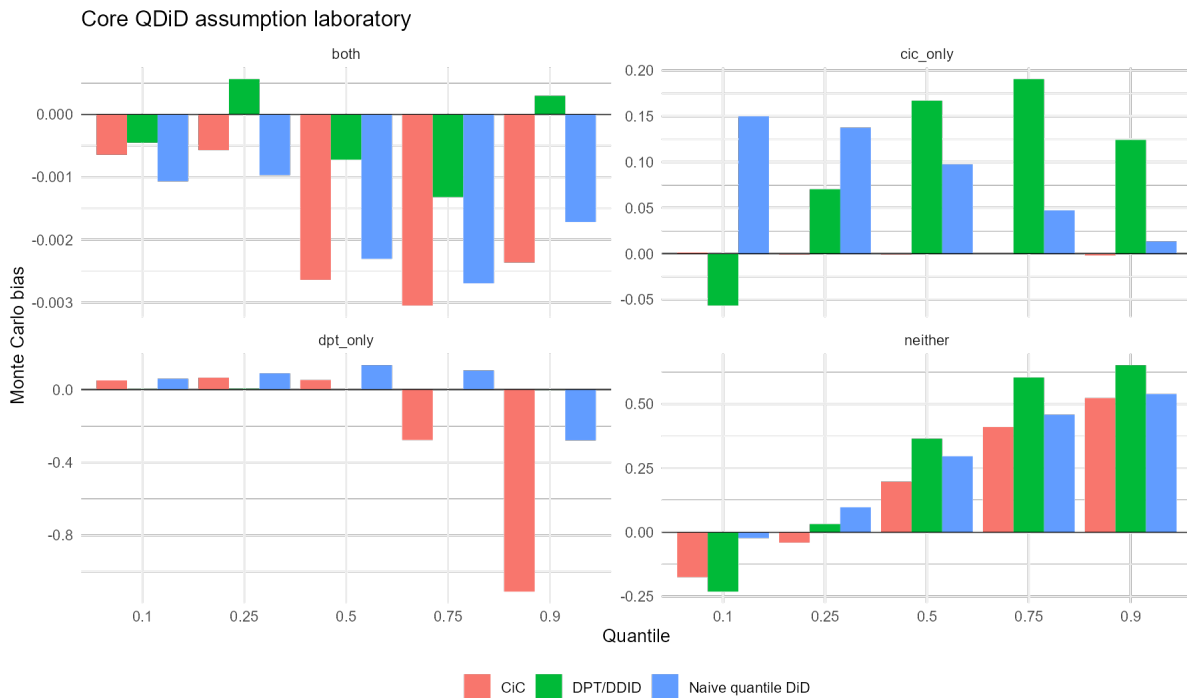


Figure 3: Core assumption laboratory. Bars report bias for naive quantile DiD, CiC, and DPT/DDID over five quantiles at 4,000 observations per cell (1,000 replications). Each estimator performs well when its identifying restriction holds; misspecification produces quantile-specific bias. QTT denotes a difference of marginal quantiles.

12.2 Lesson 2: Misspecification persists as the sample grows

The bias of an estimator built on the wrong counterfactual restriction does not disappear with a larger sample. In the panel module, mean RMSE falls from 0.174 at $N = 500$ to 0.122 at $N = 1,500$ when both transition restrictions hold, but remains 0.451 when the untreated-change restriction fails and 1.078

when the copula restriction fails. Doubly robust estimation similarly has RMSE 0.102–0.104 when either nuisance model is correct and 0.449 when both are wrong; this robustness is conditional on the causal design. The specialized modules carry the same lesson to tails, compliance, multivariate outcomes, and staggered aggregation, but they target different estimands and are therefore summarized only in Appendix C. Shape correction or rearrangement can regularize a noisy estimated CDF; it cannot repair a systematically false identifying restriction.

12.3 Lesson 3: Uniform inference reveals process-wide failure

When the maintained restriction is valid, uniform coverage is 0.92–0.93 for CiC and 0.91–0.97 for DPT (Monte Carlo standard error at most 0.029). Under misspecification, CiC coverage falls to 0.24 in the DPT-only design and 0.01 when neither restriction holds; DPT coverage falls to 0.30 in the CiC-only design and 0 when neither holds. Pointwise intervals can look less alarming because they need not cover the entire inspected quantile process.

The simultaneous pretrend diagnostic is a pedagogical implementation of established empirical-process and multiplier-bootstrap methods, not a new test (Chernozhukov et al., 2017, 2020, 2019). The caution against treating pretrend non-rejection as design validation also parallels the mean-DiD evidence in Roth (2022). Its null rejection rate is 0.070 at 250 observations per cell (MCSE 0.011), a mild size distortion relative to the nominal 0.05 level, and 0.054 at 1,000 (MCSE 0.010). Against the largest persistent violation it rises to 0.246 (MCSE 0.019) and 0.738 (MCSE 0.020). These results show that a diagnostic can contradict observable implications or reveal instability; non-rejection cannot validate a post-treatment counterfactual.

12.4 Lesson 4: Resampling must follow assignment

When treatment is assigned by cluster, unit resampling treats correlated observations as independent. In the application-calibrated design with 11 treated and 20 comparison clusters, uniform coverage is 0.77 under unit resampling (MCSE 0.042) and 0.94 under assignment-cluster resampling (MCSE 0.024). In the 15/15 and 30/30 benchmarks, cluster-bootstrap coverage is 0.92 and 0.93, compared with 0.78 and 0.80 under unit resampling. The calibrated result supports assignment-level resampling in this DGP but does not guarantee validity for the empirical application; few treated clusters remain a limitation. Figure 4 combines this result with the diagnostic experiment.

The simulations reinforce one decision rule: choose a method because its counterfactual restriction is defensible, not because it wins a generic performance comparison. Larger samples sharpen sampling uncertainty; they do not choose among CiC, DPT, dependence restrictions, or bounds.

13 Conclusion: Choosing a Method in Practice

QDiD is a design problem before it is an estimation problem. The analysis begins with the policy-relevant feature of the distribution and the population to which it refers. A threshold probability, a difference of

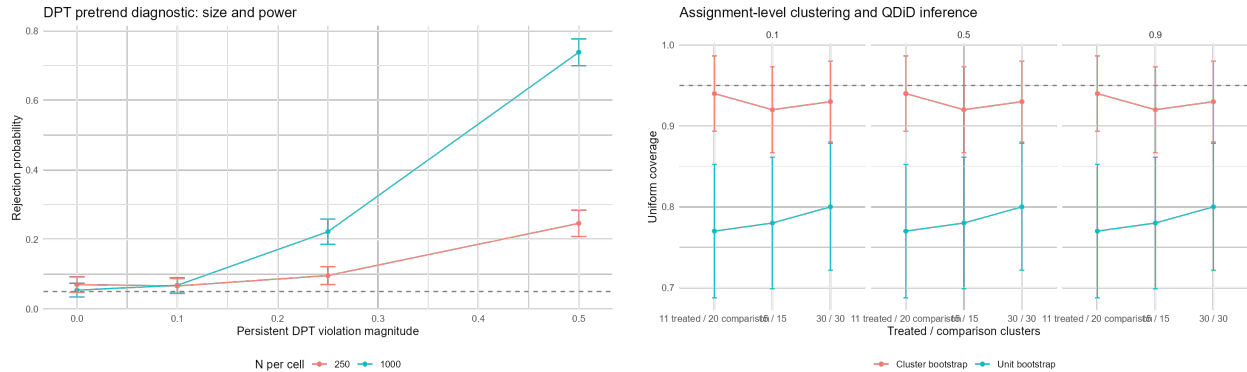


Figure 4: Diagnostic and inference experiments. Left: rejection rates for the illustrative simultaneous DPT pretrend diagnostic. Right: uniform coverage for QTTs (differences of marginal quantiles) under unit and assignment-cluster resampling. Error bars are 95% Monte Carlo intervals. The diagnostic applies existing simultaneous-inference methods; it is not a new test.

marginal quantiles, a tail risk, a joint distributional effect, and an identified set are not interchangeable targets. The available data then determine which counterfactual stories can even be considered: repeated cross-sections reveal marginal distributions, linked panels reveal transitions, and staggered timing creates a collection of cohort-time problems rather than one treated-post comparison.

The decisive step is to state how the missing treated-post untreated distribution is constructed. Stable ranks lead toward CiC; parallel changes in threshold probabilities lead toward DPT; observed histories can support a transition or dependence model; and weaker knowledge leads naturally to bounds. Estimation should follow this argument rather than precede it. The same principle governs diagnostics: support and placebo transports speak to CiC, threshold processes and raw-CDF shape speak to DPT, and transition stability and attrition speak to panel methods. None of these checks proves the unobserved post-treatment restriction, but they show the reader where the extrapolation lies.

Finally, inference should cover the object readers actually inspect and should respect the level of treatment assignment. A quantile curve calls for a simultaneous band, and a state-level policy calls for state-level resampling. A competing counterfactual, a sensitivity analysis, or bounds should remain visible when the institutional argument does not clearly select one model. The master guide in Table 2 collects these choices, but it is not an algorithm. CiC, DPT, panel restrictions, staggered estimators, and specialized extensions answer different counterfactual questions. Transparent assumptions and a cautious identified set are more informative than a precise estimate built on an indefensible restriction.

A Literature Inclusion and Classification Audit

The review searched material available through July 21, 2026. Keyword combinations—“quantile difference-in-differences,” “distributional difference-in-differences,” “changes-in-changes,” “panel quantile treatment effects,” “distributional parallel trends,” “fuzzy changes-in-changes,” and “extreme changes-in-changes”—were checked in Google Scholar, RePEc/IDEAS, NBER, IZA, arXiv, and publisher pages. Backward references

from the canonical papers and forward citations to them were then screened. This search-and-screen process is a defensible starting point, not a systematic review: it was not conducted under a PRISMA-style protocol with dual independent screening, and a reader who needs that stronger guarantee should treat the figures below as describing this paper’s retained review set rather than a full census of search hits. The reviewed bibliography contains 72 unique works. Fifty enter the method map used to build the paper’s tables—44 peer-reviewed articles and six working papers—while 22 are retained only as contextual applications, general econometric foundations, or neighboring results. Each working paper is flagged as such at first citation rather than silently treated as a settled result, and a published version is cited in place of an earlier working-paper draft wherever one was located. The final check verified titles, versions, publication status, software links, and citation metadata against publisher, journal, CRAN, SSC, or author pages where available.

A work enters the core map if it identifies a DiD counterfactual distribution or distributional functional; changes the data, support, compliance, timing, or dependence restrictions needed for that object; develops inference directly used for distributional counterfactuals; or supplies a mean-DiD benchmark necessary to define the survey boundary. Generic cross-sectional QTE papers and applications enter only as context. The machine-readable audit in `Programs/ReplicationV18/literature_map.csv` records family, estimand, data structure, role, publication status, and inclusion reason. Working papers are labeled as evolving contributions rather than treated as settled results.

Table 9: Auditable map from the core literature to the review taxonomy

Family	Representative work	Primary object	Status	Reason included
Rank transport	Athey and Imbens (2006); Melly and Santangelo (2015)	Counterfactual CDF, QTT, CQTT	Published; working paper	Canonical CiC and covariate extension
Panel dependence	Callaway and Li (2019); Callaway et al. (2018)	Panel and repeated-CS QTT	Published	Makes longitudinal dependence part of identification
Distributional trends	Bonhomme and Sauder (2011); Kim and Wooldridge (2025); Fernández-Val et al. (2025)	DTT, QTT, joint CDF	Published; working paper	Threshold- and link-scale distributional DiD
Staggered distributional/QTT	Li and Lin (2024); Ciaccio (2025); Galvao and Miller (2026)	Cohort-time QTT under staggered timing	Published; working papers	Extends copula-invariant DPT to staggered adoption, adds a plug-in estimator, and adds a doubly robust estimator
Partial identification	Fan and Yu (2012); Rothe (2012)	Counterfactual CDF and QTT bounds	Published	Defines the branch without dependence point identification
Specialized targets	Sasaki and Wang (2024); de Chaisemartin and D’Haultfoeuille (2018)	Tail and complier effects	Published	Adds tail regularity or compliance restrictions
Process inference	Chernozhukov et al. (2013); Chernozhukov et al. (2020); Donald and Hsu (2014)	CDF/quantile bands	Published	Supports uniform inference and inversion
Mean-DiD boundary	Callaway and Sant’Anna (2021); Roth et al. (2023); Baker et al. (2026)	ATT and cohort-time ATT	Published	Establishes the complementary modern-DiD boundary

B Supplementary Estimation Details

B.1 Threshold-DPT implementation used in the application

The empirical illustration deliberately implements DPT through the threshold operator rather than through additive changes in marginal quantiles. Let $\hat{F}_{gt}(y)$ denote the empirical CDF for group g and period t on a stated threshold grid $\mathcal{Y}$. The raw treated-post untreated counterfactual is

$$\hat{F}_{11}^{0,raw}(y) = \hat{F}_{10}(y) + \hat{F}_{01}(y) - \hat{F}_{00}(y), \quad y \in \mathcal{Y}.$$

This process is stored before any correction. The replication code records its minimum, maximum, number of downward steps, and largest downward step. Those quantities distinguish small empirical irregularities from a counterfactual operator that is systematically incompatible with a CDF.

The shape decision uses a data-adaptive grid of 99 empirical outcome quantiles selected once from the original sample and held fixed during resampling. States are resampled separately within the treated and comparison groups. For each of 999 draws, the four cell CDFs and the raw DPT operator are reconstructed. A studentized supremum critical value produces a simultaneous band for $F_{11}^{0,raw}(y)$. Feasibility is checked by asking whether there exists a nondecreasing function in $[0, 1]$ lying between the lower and upper band at every reported threshold. In the application this condition is satisfied. The point operator is then clipped to $[0, 1]$ and subjected to a cumulative-maximum shape correction. The corrected counterfactual quantile is

$$\hat{Q}_{11}^{0,DPT}(\tau) = \inf\{y : \hat{F}_{11}^{0,corr}(y) \geq \tau\},$$

and the reported effect is $\hat{Q}_{11}^1(\tau) - \hat{Q}_{11}^{0,DPT}(\tau)$.

The QTT band repeats the cumulative-maximum correction and inversion inside every state-stratified bootstrap draw. Its critical value is based on the supremum of the studentized QTT process over $\tau \in \{0.1, \dots, 0.9\}$. Thus the raw-CDF band answers whether numerical shape regularization is defensible, while the QTT band describes sampling uncertainty conditional on DPT. Neither band tests the unobserved post-treatment parallel-trends restriction.

This sequence differs from the additive quantile-change estimator

$$\hat{Q}_{10}(\tau) + \hat{Q}_{01}(\tau) - \hat{Q}_{00}(\tau).$$

That expression may be useful under a quantile-trend restriction, but it is not obtained by inverting the DPT CDF operator. The V18 archive keeps the two constructions separate in filenames, stored objects, tables, and figures. The implementation audit therefore has four reproducible checkpoints:

1. a fixed threshold and quantile grid;
2. the uncorrected DPT CDF and its simultaneous band;
3. the stated shape-correction and inversion rule; and

4. assignment-cluster resampling of the complete estimator.

These checkpoints are useful beyond this application because they reveal where identification, numerical shape enforcement, and sampling uncertainty enter the reported curve.

The main text retains the counterfactual identification formulas and moves secondary plug-in and bootstrap algebra here.

B.2 Panel QTT implementation

Estimation mirrors these identification formulas by replacing population cdfs with their empirical counterparts (possibly after reweighting). For example, under unconditional assumptions an estimator for $F_{Y_{0t}|D=1}(y)$ is

$$\hat{F}_{Y_{0t}|D=1}(y) = \frac{1}{n_D} \sum_{i \in \mathcal{D}} \mathbf{1} \left\{ \hat{F}_{\Delta Y_t|D=0}^{-1} \left(\hat{F}_{\Delta Y_{t-1}|D=1}(\Delta Y_{i,t-1}) \right) \leq y - \hat{F}_{Y_{t-1}|D=1}^{-1} \left(\hat{F}_{Y_{t-2}|D=1}(Y_{i,t-2}) \right) \right\},$$

where n_D is the number of treated units and $\hat{F}$ denotes empirical cdfs. With covariates, the reweighted cdf

$$\hat{F}_{\Delta Y_{0t}|D=1}^p(\delta) = \frac{\frac{1}{n} \sum_{i=1}^n (1 - D_i) \frac{\hat{p}(X_i)}{1 - \hat{p}(X_i)} \mathbf{1}\{\Delta Y_{it} \leq \delta\}}{\frac{1}{n} \sum_{i=1}^n (1 - D_i) \frac{\hat{p}(X_i)}{1 - \hat{p}(X_i)}}$$

is used to approximate $F_{\Delta Y_{0t}|D=1}^p(\cdot)$, where $\hat{p}(X)$ is a first-stage estimate of the propensity score. The estimated counterfactual quantiles $\hat{F}_{Y_{0t}|D=1}^{-1}(\tau)$ and treated quantiles $\hat{F}_{Y_{1t}|D=1}^{-1}(\tau)$ then yield $\widehat{QTT}(\tau)$

For inference, Callaway and Li (2019) show that the QTT process converges weakly to a Gaussian process at the usual $\sqrt{n}$ rate and propose an empirical bootstrap that yields *uniform* confidence bands over $\tau \in \mathcal{T}$. Let $\widehat{QTT}^b(\tau)$ denote the QTT estimate in bootstrap replication b and let $\hat{\Sigma}(\tau)$ be the estimated variance function obtained by a studentized interquartile-range scaling. For each $b = 1, \dots, B$ define

$$I^b = \sup_{\tau \in \mathcal{T}} \hat{\Sigma}(\tau)^{-1/2} \left| \sqrt{n} (\widehat{QTT}^b(\tau) - \widehat{QTT}(\tau)) \right|.$$

If $c_{1-\alpha}^B$ denotes the $(1 - \alpha)$ quantile of $\{I^b\}_{b=1}^B$, a $(1 - \alpha)$ uniform confidence band is

$$\hat{C}_{QTT}(\tau) = \widehat{QTT}(\tau) \pm c_{1-\alpha}^B \hat{\Sigma}(\tau)^{1/2} / \sqrt{n}, \quad \tau \in \mathcal{T}.$$

The studentization makes the bootstrap approximation approximately pivotal in finite samples, which is especially useful when one reports QTT curves over a grid of quantiles.

Despite these attractive features, the Callaway and Li (2019) approach also has limitations that applied researchers must weigh carefully. Data requirements are stronger than in standard DiD: one needs panel data with *at least* two pre-treatment periods; repeated cross-sections are insufficient. The Copula Stability Assumption is conceptually intuitive but substantively strong and fundamentally untestable, and may be questionable in settings with evolving sorting or changes in how shocks propagate over time. The framework

also assumes continuously distributed outcomes, which can complicate applications with discrete or mass-point outcomes. In practice, these methods are most compelling when rich panel data are available, pre-trends appear stable, and there are substantive reasons to believe that the dependence structure between levels and changes of untreated outcomes does not shift over time.

B.3 Repeated-cross-section CQTT implementation

Estimation proceeds nonparametrically, exploiting the discreteness of X . For each $x \in \mathcal{X}$, group $d \in \{0, 1\}$, and period $s \in \{t-1, t\}$, estimate the conditional CDF from the corresponding repeated-cross-section cell,

$$\hat{F}_{Y_s|X=x, D_t=d}(y) = \frac{1}{n_{sxd}} \sum_{i: S_i=s} \mathbf{1}\{Y_i \leq y, X_i = x, D_i = d\},$$

where n_{sxd} is the number of observations in that cell. The treated-post conditional distribution is observed directly. The counterfactual requires the additional conditional rank-invariance restriction because repeated cross-sections contain no observed ΔY_i .

For each baseline comparison observation i with $(S_i, D_i, X_i) = (t-1, 0, x)$, compute its estimated conditional rank

$$\hat{u}_i = \hat{F}_{Y_{t-1}|X=x, D=0}(Y_i),$$

the rank-matched comparison-group pseudo-change

$$\widehat{\Delta^g Y}_i = \hat{Q}_{Y_t|X=x, D=0}(\hat{u}_i) - Y_i,$$

and the treated baseline outcome at the same rank,

$$\hat{Y}_i^{T0} = \hat{Q}_{Y_{t-1}|X=x, D=1}(\hat{u}_i).$$

The conditional counterfactual CDF is the empirical distribution of their sum,

$$\hat{F}_{Y_t(0)|X=x, D_t=1}(y) = \frac{1}{n_{t-1,x,0}} \sum_{i: S_i=t-1} \mathbf{1}\left\{\widehat{\Delta^g Y}_i + \hat{Y}_i^{T0} \leq y, X_i = x, D_i = 0\right\}.$$

This is the sample analogue of the repeated-cross-section corollary in Callaway et al. (2018). It never substitutes an unobserved individual change. Conditional quantiles are obtained by inversion, and the CQTT subtracts the counterfactual quantile from the observed treated-post quantile. Unconditional QTTs require first averaging the conditional counterfactual CDF over the treated target distribution of X and then inverting; averaging conditional quantiles would generally estimate a different object.

Callaway et al. (2018) show, using empirical process techniques, that the CQTT process $\{\hat{\Delta}_x^{QTT}(\tau)\}$ converges in distribution to a Gaussian process at the parametric rate $\sqrt{n}$. However, the limiting distribution is not nuisance-parameter free: it reflects first-step estimation error in the conditional cdfs and the complexity of the identification mapping. In practice, this motivates the use of bootstrap methods to obtain standard

errors and confidence bands for CQTT across τ and x .

From an applied perspective, the Callaway et al. (2018) approach is particularly useful when only two periods are available and covariates are naturally discrete (for example, education categories, race, or region). The data distinction must remain visible: with a two-period panel, observed comparison changes support cross-group copula invariance; with repeated cross-sections, conditional rank invariance creates pseudo-changes from marginal rank maps. Fine discrete cells can be sparse, continuous covariates must be modeled or coarsened, and the design remains two-period and sharp-treatment. Dynamic or staggered settings require other methods.

B.4 Covariate-adjusted DPT via inverse probability weighting

The plug-in estimator for the Kim and Wooldridge (2025) construction of Section 7.2 replaces cell probabilities with first-stage propensity estimates and reweights the additive DPT operator observation by observation rather than integrating conditional CDFs cell by cell. Writing $\mathbf{1}_{i,gt} = \mathbf{1}\{G_i = g, T_i = t\}$ for cell indicators and $\hat{p}_{g,t}(x)$ for an estimated cell propensity (for example, a multinomial logistic series estimator),

$$\hat{F}_{11}^0(y) = \frac{1}{n} \sum_{i=1}^n \frac{\hat{p}_{11}(X_i)}{\hat{p}_{11}} \left\{ \frac{\mathbf{1}_{i,10}}{\hat{p}_{10}(X_i)} + \frac{\mathbf{1}_{i,01}}{\hat{p}_{01}(X_i)} - \frac{\mathbf{1}_{i,00}}{\hat{p}_{00}(X_i)} \right\} \mathbf{1}\{Y_i \leq y\},$$

with the observed treated-post CDF estimated directly as $\hat{F}_{11}^1(y) = \frac{1}{np_{11}} \sum_i \mathbf{1}_{i,11} \mathbf{1}\{Y_i \leq y\}$. Inverting both and differencing gives $\widehat{QTT}(\tau)$.

Because the operator is additive, $\hat{F}_{11}^0$ can leave $[0, 1]$ or fail monotonicity for the same reason the unconditional raw DPT operator can (Appendix B’s threshold-DPT implementation). Kim and Wooldridge test this shape restriction directly: over a threshold grid $\mathcal{Y}$ that may grow with the sample, they form moment inequalities comparing $\hat{F}_{11}^0$ at every pair of grid points—not only adjacent ones, which increases power without altering the null—and control size using the high-dimensional bootstrap bounds of Chernozhukov et al. (2017) together with an analytical rather than bootstrapped variance estimate. Rejecting the test is evidence against the DDID restriction on the reported grid, not a numerical defect to be clipped away. Inference for the CDF, quantile, and QTT processes uses the generic bootstrap-and-monotonize recipe already used throughout this appendix: resample, monotone each bootstrap CDF, invert to quantile functions, and take the studentized supremum over the bootstrap draws to build simultaneous bands, a route that accommodates discrete or mixed outcomes without a separate discrete-outcome correction.

B.5 Cohort-time QTT via copula invariance across groups

Li and Lin (2024) identify the cohort-time counterfactual $F_{g,t}^0$ (Section 8.3) but do not propose an estimator; Ciaccio (2025) supplies the sample analogue used here. Let $C_i = 1$ index never-treated units, let $\rho \geq 0$ be a limited-anticipation parameter, and let $\Delta_{[g-\rho-1,t]}Y = Y_t - Y_{g-\rho-1}$ denote the long difference between the period just before cohort g adopts (adjusted for anticipation) and calendar period t . The plug-in estimator

of the cohort-time counterfactual CDF is

$$\hat{F}_{g,t}^0(y) = \frac{1}{n_0} \sum_{i:C_i=1} \mathbf{1} \left\{ \hat{F}_{\Delta Y|C=1}^{-1} \left(\hat{F}_{\Delta Y|C=1}(\Delta_{[g-\rho-1,t]} Y_i) \right) \leq y - \hat{F}_{Y_{g-\rho-1}|G=g}^{-1} \left(\hat{F}_{Y_{g-\rho-1}|C=1}(Y_{i,g-\rho-1}) \right) \right\},$$

where n_0 is the number of never-treated units observed in periods $g-\rho-1$ and t ; the not-yet-treated analogue replaces $C = 1$ throughout with eligibility through some later period s . Inverting yields $\hat{F}_{g,t}^{-1,0}(\tau)$, and with $\hat{F}_{g,t}^1$ the empirical CDF of cohort g 's observed outcomes at t ,

$$\widehat{QTT}_{g,t}(\tau) = \hat{F}_{g,t}^{-1,1}(\tau) - \hat{F}_{g,t}^{-1,0}(\tau).$$

Inference repeats the studentized supremum construction of the Panel QTT implementation above, with bootstrap replications $\widehat{QTT}_{g,t}^b(\tau)$ resampled at the assignment level in place of $\widehat{QTT}_{g,t}(\tau)$; the same interquartile-range studentization keeps the approximation approximately pivotal. Aggregation follows by plugging cohort-time estimates into sample-analogue weights, for example the event-time average

$$\hat{\theta}^e(\tau) = \sum_g \mathbf{1}\{g + e \leq T\} \widehat{\Pr}(G = g \mid g + e \leq T) \widehat{QTT}_{g,g+e}(\tau),$$

which reports the effect of e periods of exposure by averaging cohort-specific quantile contrasts at the same event time rather than mixing cohort-time distributions first. When rank invariance between cohort g 's treated and counterfactual outcomes is doubted, the same $\hat{F}_{g,t}^0$ and $\hat{F}_{g,t}^1$ feed a stochastic-dominance test in place of $\widehat{QTT}_{g,t}(\tau)$, following Maasoumi and Wang (2019), with critical values from a pair bootstrap over units.

B.6 Doubly robust cohort-time QTT

Galvao and Miller (2026) replace the plain empirical $\hat{F}_{\Delta Y|C=1}$ in the construction above with a doubly robust estimate of the same long-difference distribution. Let $\hat{p}_{g,s}(x)$ denote an estimated generalized propensity score for cohort g against a not-yet-treated-by- s comparison group, and let $\hat{P}_{g,s}(\delta \mid x)$ denote an estimated model for the conditional CDF of the long difference given $X = x$ among that comparison group (a sieve-logit propensity and a kernel or parametric conditional-CDF model are typical choices; neither is cross-fit). The doubly robust estimator of $F_{\Delta Y(0)|G=g}$ is

$$\hat{F}_{\Delta Y(0)|g}(y) = \frac{1}{n} \sum_{i=1}^n \left[\frac{(1 - D_{is})(1 - \mathbf{1}_{ig})}{\widehat{\Pr}(G = g)} \cdot \frac{\hat{p}_{g,s}(X_i)}{1 - \hat{p}_{g,s}(X_i)} \mathbf{1}\{\Delta Y_i \leq y\} - \left(\frac{(1 - D_{is})(1 - \mathbf{1}_{ig})}{\widehat{\Pr}(G = g)} \cdot \frac{\hat{p}_{g,s}(X_i)}{1 - \hat{p}_{g,s}(X_i)} - \frac{\mathbf{1}_{ig}}{\widehat{\Pr}(G = g)} \right) \hat{P}_{g,s} \right]$$

which remains consistent for $F_{\Delta Y(0)|G=g}$ if either $\hat{p}_{g,s}$ or $\hat{P}_{g,s}$ is correctly specified, not necessarily both; the never-treated version replaces the not-yet-treated indicator with the never-treated indicator throughout, using the same weighting scheme Callaway and Sant'Anna (2021) use for the mean ATT(g, t). Substituting $\hat{F}_{\Delta Y(0)|g}$ for $\hat{F}_{\Delta Y|C=1}$ in the rank-composition formula of the previous appendix subsection yields the doubly

robust counterfactual $\widehat{F}_{g,t}^0$, and inversion and differencing yield $\widehat{QTT}_{g,t}(\tau)$ as before.

Both nuisance components are estimated on the full sample and held to a uniform $o_p(n^{-1/4})$ rate condition, the classical semiparametric two-step requirement rather than the cross-fitted rate relaxation of Section 9.2’s machine-learning discussion; the source paper shows this rate condition is met by correctly specified parametric models and by sieve-logit/kernel nonparametric estimators under standard smoothness conditions. Inference uses a nonparametric bootstrap over units—resampling the full panel, re-estimating both nuisance components, and reconstructing $\widehat{QTT}_{g,t}(\tau)$ in every draw—rather than a closed-form asymptotic variance, because the limiting Gaussian process depends on the nuisance functions in a way that has no simple analytic form. The source paper notes explicitly that the usual double-robustness relaxation of the required nuisance convergence rate, which can apply to point estimation, need not carry over to bootstrap-based inference; this is why both nuisance rates are held at $o_p(n^{-1/4})$ rather than relaxed.

C Monte Carlo Designs and Estimator Implementation

All production experiments use seed 20260720 and fixed oracle populations of 1.5 million observations. Table 10 gives the replication counts. Replication-level estimates, failures, CDF-validity indicators, coverage indicators, and all summary inputs are stored as CSV files. Bias Monte Carlo standard errors are $sd(\widehat{\theta} - \theta)/\sqrt{R}$; RMSE standard errors use the delta method applied to the mean squared error. Coverage and rejection-probability standard errors are binomial Monte Carlo standard errors.

Table 10: Production Monte Carlo configuration

Module	MC replications	Bootstrap draws	Sample sizes
Core continuous and discrete	1,000/design	299; 100 reps at $n = 1,000$	250, 1,000, 4,000/cell
Diagnostic size and power	500/design	299	250 and 1,000/cell
Clustered inference	250/design	299; 100 reps	11 treated/20 comparison; 15/15; 30/30; 40 units/cluster
Panel QTT	300/design	—	500 and 1,500 individuals
DR, tail, staggered, fuzzy, multi-variate	500/design	—	1,500 unless stated below

Core assumption laboratory. Let $k(z) = 0.35 + 1.15z + 0.10 \tanh(z)$. In the “both” design, $Z \sim N(0, 1)$ in every group, $Y_0 = Z$ before treatment, and $Y_0 = k(Z)$ afterward. In the CiC-only design, $U \sim U(0, 1)$ for comparisons and $U \sim \text{Beta}(2.2, 1.3)$ for treated units, $Z = \Phi^{-1}(U)$, and the same k maps ranks into post-period outcomes. In the DPT-only design,

$$F_{gt} = (1 - w_{gt})A + w_{gt}B, \quad w_{gt} = 0.30g + 0.25t,$$

where $A = N(-0.35, 0.85^2)$ and $B = N(1.10, 1.20^2)$; hence $F_{11} - F_{10} = F_{01} - F_{00}$ exactly. The “neither” design starts from the CiC-only design and adds $0.45 \tanh(Z) + 0.20Z^2/(1 + Z^2)$ only to the treated-post counterfactual. Treated-post observed outcomes add $0.55 - 0.35\Phi\{\text{std}(Y_0)\}$. The discrete design rounds both potential and observed outcomes to the nearest one-half.

Table 11: Selected outputs from representative Monte Carlo designs

Module/design	Restriction status	Sample; R	Target; truth	Bias (MCSE)	RMSE (MCSE)	Uniform coverage (MCSE)	CDF failures
Core CiC; both valid	Valid	1,000/cell; 1,000	QTT(.50); .375	-.001 (.003)	.100 (.002)	.930 (.026)	—
Core DPT; DPT false	False	4,000/cell; 1,000	QTT(.50); .376	.167 (.001)	.170 (.001)	—	—
Panel; both valid	Valid	1,500; 300	QTT(.50); .330	.000 (.004)	.075 (.003)	—	—
DR; propensity only correct	Valid via DR	1,500; 500	QTT(.50); .338	-.025 (.003)	.078 (.003)	—	—
Extreme CiC; Pareto, $k = 39$	Valid; difficult tail	1,500; 500	QTT(.99); .500	-.167 (.061)	1.362 (.098)	—	—
Fuzzy; 10% compliers	Valid; weak first stage	1,500; 500	CQTT(.50); .425	.205 (.033)	.759 (.024)	—	.956
Staggered; opaque pool	Pooling invalid	1,500/cohort; 500	QTT(.50); .228	.119 (.002)	.125 (.002)	—	—
Joint DRegDiD; copula false	Joint restriction false	1,500/cell; 500	DTT; -.044	-.123 (.001)	.125 (.001)	—	.000
Cluster bootstrap; 11/20	Valid DPT DGP	40/cluster; 250	QTT(.50); .340	.008 (.012)	.192 (.007)	.940 (.024)	—

Notes: Each row is a prespecified representative design, not a cross-module ranking; targets, samples, and inferential objects differ across modules. The restriction-status column states whether the relevant identifying or aggregation restriction holds in that row. Coverage uses 100 inference replications with 299 bootstrap draws where reported and is interpreted illustratively at the displayed Monte Carlo precision. “—” means that coverage or CDF validity is not an output of that module, not that performance is perfect. The fuzzy invalid-CDF rate concerns raw complier mixtures. Full quantile/threshold results, denominators, and failure indicators are in `mc_standardized_reporting.csv` and the replication-level files.

The estimators are the difference in marginal quantile changes, canonical empirical CiC, and the raw DPT operator $\hat{F}_{10} + \hat{F}_{01} - \hat{F}_{00}$ followed by cumulative-maximum shape correction and inversion. A raw DPT CDF is flagged when it leaves $[-0.02, 1.02]$ or has a downward increment below -0.02 . Bootstrap inference resamples the four cells independently. Pointwise percentile intervals and studentized supremum bands over the five-quantile process are both evaluated.

Local violations and diagnostic power. The continuous violation path adds

$$\lambda \left\{ 0.50 \tanh(Z) + \frac{0.15Z^2}{1 + Z^2} \right\}$$

to the treated-post counterfactual for $\lambda \in \{0, .2, .4, .6, .8, 1\}$. The diagnostic experiment uses the valid DPT mixture above in two pretreatment transitions and adds $\lambda\{0.30 + 0.35 \tanh(Y_0)\}$ to the treated group’s second transition for $\lambda \in \{0, .10, .25, .50\}$. It evaluates the multiplier-bootstrap supremum statistic defined in Section 10 on nine pooled-quantile thresholds. Rejection at $\lambda = 0$ estimates size; rejection for $\lambda > 0$ estimates power against persistent violations. These experiments cannot detect violations that arise only after treatment, which is why non-rejection is not interpreted as validation.

Clustered treatment. Clusters are assigned to treated or comparison groups and observed in both periods. Outcomes contain a cluster intercept with standard deviation 0.60, a cluster-period shock with standard deviation 0.35, a unit component, and a common time shift of 0.20. The treated-post effect is $0.45 - 0.20\Phi\{Y_0/(1 + 0.60^2 + 0.35^2)^{1/2}\}$. DPT holds in population. Uniform coverage is computed once by resampling units within group-period cells and once by resampling entire assignment clusters within treatment groups.

Panel dependence. Three-period panels generate a baseline level, a first untreated change $0.20 + 0.35Y_{-1} + \varepsilon_0$, and a second change $0.20 + 0.35Y_0 + \varepsilon_1$, with innovation standard deviation 0.65. Change stability

is violated by adding 0.45 to treated future changes. Copula stability is violated while preserving the treated change marginal by reverse-sorting future changes against treated baseline outcomes. The installed `qte::panel.qtet` implementation is called directly.

Nuisance robustness. The DR experiment draws $X \sim N(0, 1)$, treatment probability $\text{logit}^{-1}(-0.2 + 0.8X - 0.5X^2)$, untreated outcome $0.4X + 0.45X^2 + \varepsilon$, and heterogeneous effect $0.5 - 0.25\Phi(Y_0)$. Correct and misspecified propensity and outcome models omit or retain quadratic terms, yielding all four combinations required to display double robustness.

Specialized estimands. The tail module compares empirical CiC and Weissman/Hill extrapolation at $\tau \in \{.90, .95, .975, .99\}$ under Pareto and lognormal tails and thresholds $k \in \{\lfloor \sqrt{n} \rfloor, \lfloor 2\sqrt{n} \rfloor\}$. The staggered module has cohorts adopting in periods 2, 3, and 4 plus a never-treated group; cohort effects differ in sign and grow with event time, so an opaque pooled contrast has no reason to recover the cohort-time targets. The fuzzy module varies complier shares 0.10, 0.30, and 0.60 and adds a defier share of zero or 0.08. The multivariate module varies dimension one through three and changes the treated-post Gaussian-copula correlation from 0.45 to -0.45 while preserving marginal shifts. These modules target their own causal objects and are not entered into the core estimator horse race.

D Empirical Computation and Inferential Target

The main point estimators use the balanced 2005–2007 panel of 2,360 counties. State-cluster resampling draws the 11 treated and 20 comparison states separately with replacement, retains all counties and years in a selected state, creates unique bootstrap panel identifiers when a state is selected more than once, and re-estimates the entire quantile process. Main process bands and both Fan–Yu bound endpoints use 999 draws. Pretreatment placebo bands use 499 draws and control simultaneously over years 2002–2006 and $\tau \in \{.1, .5, .9\}$ within each method. The leave-one-state-out analysis deletes each of the 31 states once and is an influence diagnostic rather than an inferential procedure.

The state bootstrap is preferable to county resampling because treatment is state-level, but 31 clusters remain a limited-cluster design. Accordingly, the paper reports the number of clusters, simultaneous rather than selected-quantile uncertainty, leave-one-state-out ranges, and the wide partially identified region. The archive records seeds, bootstrap objects, session information, and machine-readable endpoint estimates. The replication manifest and one-command build are in `Programs/ReplicationV18`.

References

- Abadie, A. (2005). Semiparametric difference-in-differences estimators. *Review of Economic Studies*, 72(1):1–19.
- Abadie, A., Angrist, J., and Imbens, G. (2002). Instrumental variables estimates of the effect of subsidized training on the quantiles of trainee earnings. *Econometrica*, 70(1):91–117.
- Angrist, J., Chernozhukov, V., and Fernández-Val, I. (2006). Quantile regression under misspecification, with an application to the u.s. wage structure. *Econometrica*, 74(2):539–563.
- Athey, S. and Imbens, G. W. (2006). Identification and inference in nonlinear difference-in-differences models. *Econometrica*, 74(2):431–497.
- Baker, A., Callaway, B., Cunningham, S., Goodman-Bacon, A., and Sant’Anna, P. H. C. (2026). Difference-in-differences designs: A practitioner’s guide. *Journal of Economic Literature*, 64(2):498–557.
- Bitler, M. P., Gelbach, J. B., and Hoynes, H. W. (2006). What mean impacts miss: Distributional effects of welfare reform experiments. *American Economic Review*, 96(4):988–1012.
- Bonhomme, S. and Sauder, U. (2011). Recovering distributions in difference-in-differences models: A comparison of selectivity correction and nonparametric approaches. *Review of Economics and Statistics*.
- Borusyak, K., Jaravel, X., and Spiess, J. (2024). Revisiting event-study designs: Robust and efficient estimation. *Review of Economic Studies*, 91(6):3253–3285.
- Callaway, B. and Li, T. (2019). Quantile treatment effects in difference in differences models with panel data. *Quantitative Economics*, 10(4):1579–1618.
- Callaway, B., Li, T., and Oka, T. (2018). Quantile treatment effects in difference in differences models under dependence restrictions and with only two time periods. *Journal of Econometrics*, 206(2):395–413.
- Callaway, B. and Sant’Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.
- Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427.
- Cattaneo, M. D. (2010). Efficient semiparametric estimation of multi-valued treatment effects under ignorability. *Journal of Econometrics*, 155(2):138–154.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.

- Chernozhukov, V., Chetverikov, D., and Kato, K. (2017). Central limit theorems and the bootstrap in high dimensions. *Annals of Probability*, 45(4):2309–2352.
- Chernozhukov, V., Chetverikov, D., and Kato, K. (2019). Inference on causal and structural parameters using many moment inequalities. *The Review of Economic Studies*, 86(5):1867–1900.
- Chernozhukov, V., Fernández-Val, I., and Galichon, A. (2010). Quantile and probability curves without crossing. *Econometrica*, 78(3):1093–1125.
- Chernozhukov, V., Fernández-Val, I., and Melly, B. (2013). Inference on counterfactual distributions. *Econometrica*, 81(6):2205–2268.
- Chernozhukov, V., Fernández-Val, I., Melly, B., and Wüthrich, K. (2020). Generic inference on quantile and quantile effect functions for discrete outcomes. *Journal of the American Statistical Association*, 115(529):123–137.
- Ciaccio, A. (2025). Distributional difference-in-differences with multiple time periods. Working paper, Ca’ Foscari University of Venice, version of May 21, 2025, arXiv:2408.01208.
- de Chaisemartin, C. and D’Haultfoeuille, X. (2014). Fuzzy changes-in-changes. Unpublished manuscript.
- de Chaisemartin, C. and D’Haultfoeuille, X. (2018). Fuzzy differences-in-differences. *Review of Economic Studies*, 85(2):999–1028. Circulated from 2014 as an unpublished manuscript under the same title.
- de Chaisemartin, C. and D’Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996.
- de Chaisemartin, C., D’Haultfoeuille, X., and Guyonvarch, Y. (2019). Fuzzy differences-in-differences with stata. *Stata Journal*, 19(2):435–458.
- DiNardo, J., Fortin, N. M., and Lemieux, T. (1996). Labor market institutions and the distribution of wages, 1973–1992. *Econometrica*, 64(5):1001–1044.
- Doksum, K. (1974). Empirical probability plots and statistical inference for nonlinear models in the two-sample case. *The Annals of Statistics*, 2(2):267–277.
- Donald, S. G. and Hsu, Y.-C. (2014). Estimation and inference for distribution functions and quantile functions in treatment effect models. *Journal of Econometrics*, 178(3):383–397.
- Dube, A., Lester, T. W., and Reich, M. (2016). Minimum wage shocks, employment flows, and labor market frictions. *Journal of Labor Economics*, 34(3):663–704.
- Fan, Y. and Park, S. S. (2010). Sharp bounds on the distribution of treatment effects and their statistical inference. *Econometric Theory*, 26(3):931–951.

- Fan, Y. and Yu, Z. (2012). Partial identification of distributional and quantile treatment effects in difference-in-differences models. *Economics Letters*, 115(3):511–515.
- Fernández-Val, I., Meier, J., van Vuuren, A., and Vella, F. (2025). Distribution regression difference-in-differences. <https://arxiv.org/abs/2409.02311>. arXiv:2409.02311v2.
- Firpo, S. (2007). Efficient semiparametric estimation of quantile treatment effects. *Econometrica*, 75(1):259–276.
- Firpo, S., Fortin, N. M., and Lemieux, T. (2011). Decomposition methods in economics. In Ashenfelter, O. and Card, D., editors, *Handbook of Labor Economics*, volume 4A, pages 1–102. Elsevier.
- Foresi, S. and Peracchi, F. (1995). The conditional distribution of excess returns: An empirical analysis. *Journal of the American Statistical Association*, 90(430):451–466.
- Galvao, A. F. and Miller, B. I. (2026). Doubly-robust quantile treatment effects with staggered interventions. Working paper, draft dated May 9, 2026, SSRN Working Paper No. 5193241.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.
- Gunsilius, F. F. (2023). Distributional synthetic controls. *Econometrica*, 91(3):1105–1117.
- Heckman, J. J., Smith, J. A., and Clements, N. (1997). Making the most out of programme evaluations and social experiments: Accounting for heterogeneity in programme impacts. *Review of Economic Studies*, 64(4):487–535.
- Kim, D. and Wooldridge, J. M. (2025). Difference-in-differences estimator of quantile treatment effect on the treated. *Journal of Business & Economic Statistics*, 43(2):401–412.
- Koenker, R. (2005). *Quantile Regression*. Cambridge University Press, Cambridge.
- Koenker, R. and Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1):33–50.
- Lechner, M. (2011). The estimation of causal effects by difference-in-difference methods. *Foundations and Trends in Econometrics*, 4(3):165–224.
- Li, X. and Lin, L. (2024). Identification of quantile treatment effects in difference-in-differences settings with staggered adoption. *Economics Letters*, 241:111792.
- Maasoumi, E. and Wang, L. (2019). Distributional treatment effect analysis and stochastic dominance. Unpublished manuscript.
- Machado, J. A. F. and Mata, J. (2005). Counterfactual decomposition of changes in wage distributions using quantile regression. *Journal of Applied Econometrics*, 20(4):445–465.

- MacKinnon, J. G. and Webb, M. D. (2018). The wild bootstrap for few (treated) clusters. *The Econometrics Journal*, 21(2):114–135.
- Melly, B. and Santangelo, G. (2015). The changes-in-changes model with covariates. https://drive.google.com/file/d/1mYXXV3jW3E8jGpX1NSY0zjBNvX_ZHkLv/view.
- Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4(3):305–322.
- Roth, J., Sant’Anna, P. H. C., Bilinski, A., and Poe, J. (2023). What’s trending in difference-in-differences? a synthesis of the recent econometrics literature. *Journal of Econometrics*, 235(2):2218–2244.
- Rothe, C. (2010). Nonparametric estimation of distributional policy effects. *Journal of Econometrics*, 155(1):56–70.
- Rothe, C. (2012). Partial distributional policy effects. *Econometrica*, 80(5):2269–2301.
- Sant’Anna, P. H. C. and Zhao, J. (2020). Doubly robust difference-in-differences estimators. *Journal of Econometrics*, 219(1):101–122.
- Sasaki, Y. and Wang, Y. (2024). Extreme changes-in-changes. *Journal of Business & Economic Statistics*, 42(2):812–824.
- Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199.